

FIRE EFFECTS



FIRE EFFECTS – key concepts

Vegetation fires affect soils, water, air, flora, fauna, cultural resources, and human health and livelihoods.

This chapter focuses on soils, water, and flora/vegetation, due to their closer relationships to its academic context and the previous contents of the course.

The effects of fire are **the ecological changes that occur because of a fire event**. They can be direct (e.g., combustion of biomass) or indirect (e.g., changes in nutrient cycling, species composition, or landscape structure over time).

Fire is not just destructive — it can be essential for ecosystem renewal, biodiversity, and nutrient cycling. However, **changes in fire regimes** (due to climate change, land use, or fire suppression) **can turn fire from a regenerative force into a disruptive threat** (e.g., megafires, biodiversity loss).

FIRE EFFECTS - key concepts

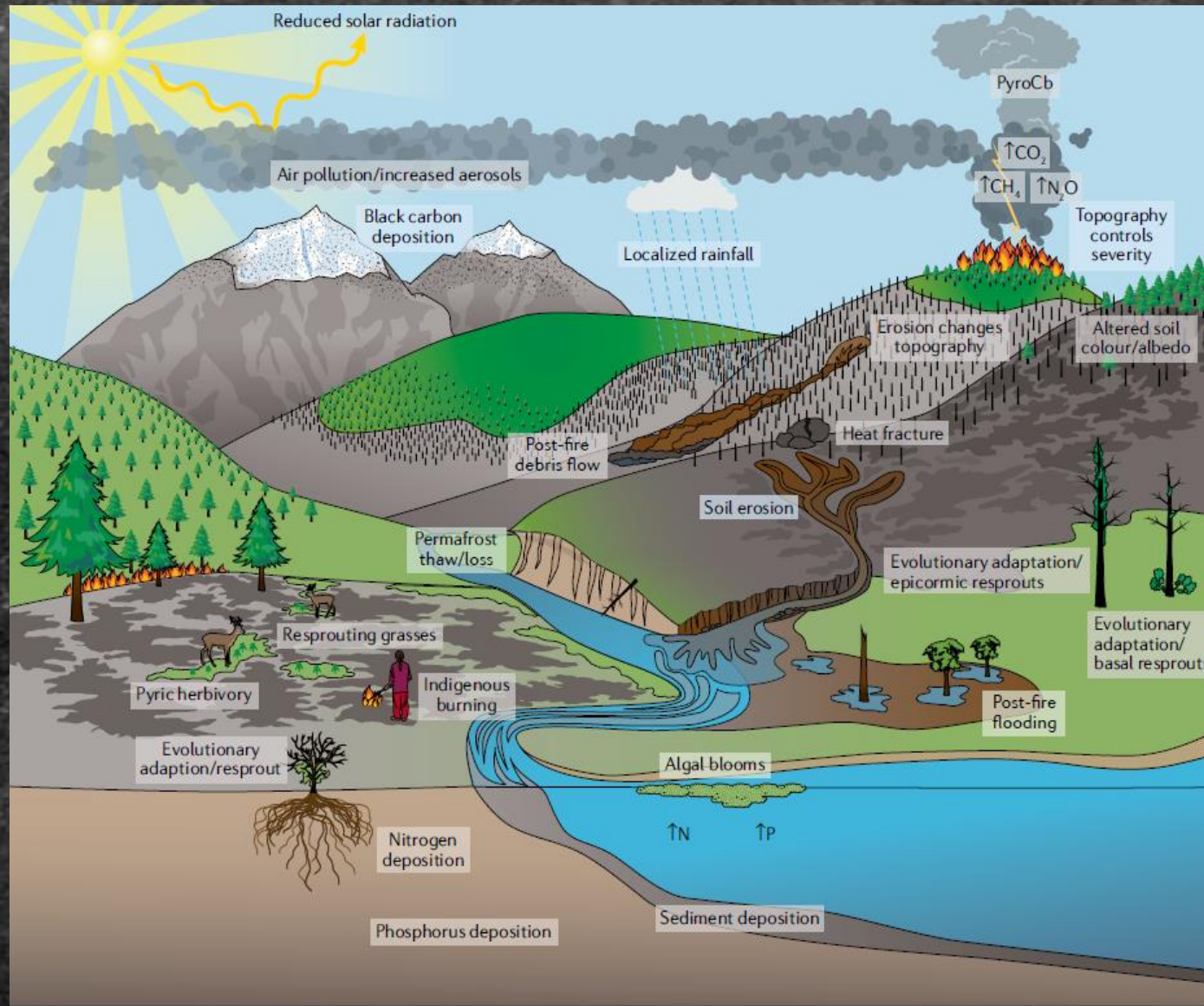


Fig. 1. Vegetation fire in the Earth system. Landscape perspective of the multiple factors that influence, interact with and are impacted by vegetation fire. **Fires have numerous direct and indirect effects that impact the biosphere (including vegetation cover), geosphere (including soil erosion), hydrosphere (including fluvial sediment and nutrient transport), cryosphere (including soot fallout and changed albedo) and atmosphere (including smoke pollution).** PyroCb, pyrocumulonimbus.

FIRE EFFECTS – key concepts

Severity is a term that describes **ecosystem responses to fire** and depends on the **quantity and rate of heat release by the fire**. It is a function of available fuel and fire behavior.

There is **not a clear relationship between fire intensity and fire severity** because the relationships between fire behavior and ecosystem responses are complex. For example, an intense, fast spreading fire may be less severe than a slow-spreading, lower intensity fire that has a longer residence time.

Severity is not quantified by a single quantitative measure of the impact of fire on an ecosystem. The relative magnitudes of fire impacts are represented on an **ordinal scale**, with classes like low / medium / high.

Fire severity classifications are **context-specific** and differ with the type of ecosystem and the resource under consideration, e.g. vegetation, fauna, soil.

FIRE EFFECTS – key concepts

Why Monitor Wildfire Effects?

1. Assess Ecosystem Response and Recovery

- Helps track vegetation regrowth, wildlife return, and soil stabilization;
- Allows identification of areas at risk of erosion, invasive species, or poor natural regeneration.

2. Inform Post-Fire Management

- Guides rehabilitation efforts, such as reforestation, erosion control, or fuel management;
- Supports decision-making for salvage logging, seed dispersal, or protection of water sources.

3. Evaluate Fire Severity and Impact

- Quantifies burn severity (e.g., vegetation and soil damage) using field data and remote sensing;
- Differentiates between low-, moderate-, and high-severity burns, which influence recovery strategies.

FIRE EFFECTS – key concepts

Why Monitor Wildfire Effects?

4. Improve Fire Prevention and Preparedness

- Lessons from past fires help refine fire behavior models and update risk maps;
- Data supports better planning of fuel treatments, such as prescribed burning or thinning.

5. Contribute to Scientific Research

- Monitoring provides essential data for understanding fire ecology, climate change feedbacks, and carbon dynamics;
- Facilitates long-term studies on resilience, succession, and landscape dynamics.

6. Support Policy, Education, and Community Awareness

- Strengthens fire management policies and land-use planning;
- Helps communicate impacts and recovery timelines to the public, stakeholders, and landowners.

FIRE EFFECTS – key concepts

Table 2. Summary of fire terminology and metrics

	Fire intensity	Fire severity	Burn severity	Ecosystem responses
Appropriate usage	Energy output from fire.	Aboveground and belowground organic matter consumption from fire.	Aboveground and belowground organic matter consumption from fire. Sometimes subdivided into 'vegetation burn severity' and 'soil burn severity'.	Functional processes that are altered by fire including regeneration, recolonization by plants and animals and watershed hydrology processes altered by fire.
Metrics	Strictly speaking it is the time-averaged energy flux in W m^{-2} , but more broadly can be measured as fireline intensity, temperature, residence time, radiant energy and other.	Aboveground measures include tree crown canopy scorch, crown volume kill, bole height scorch, skeleton twig diameter. Belowground and soil measures include ash deposition, surface organic matter, belowground organic matter contributing to soil structure, degree of hydrophobicity, and heat-induced oxidation of minerals. Mortality is a common measure that is best applied to non-sprouting trees in surface fire regimes. In crown fire regimes, aboveground mortality may be useful when fires are patchy.	Often used interchangeably with fire severity. Usually the term is applied to soils and designated 'soil burn severity'. In the USA, it is the preferred term used in post-fire Burned Area Emergency Response assessments and is considered to be the relative change due to fire, i.e. two soils with poor structure and low organic matter content may be rated differently if one was in that condition before the fire and another was not. Degree of severity may be influenced by socio-political concerns such as values at risk.	Vegetative cover, seedling recruitment, plant community composition and diversity, and plant and animal recolonization are important biotic parameters. Watershed hydrological processes such as dry ravel, erosion, and debris flows are the more important abiotic processes.
Inappropriate usage	Should never be used to describe fire effects such as those described under any of the remaining columns.	Should not include ecosystem responses. Also, in shrubland ecosystems, complete above- and belowground mortality should not be considered here because it depends on vegetation composition and the proportion of sprouting and non-sprouting species.	Should not include ecosystem responses. Also, this term should be restricted to field measurements and not be used to name remote-sensing indices because the interpretation of remote data is dependent on ground-truthing with field measurements of burn severity; calling both measures burn severity is circular.	Correlations between severity and ecosystem responses demonstrated in one system should not be considered universal for all ecosystems.

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Table 1. The matrix originally proposed by Ryan and Noste (1985) that related changes in aboveground vegetation and soil organic matter to fire severity has generally been simplified to a table such as that below; modified from Ryan (2002) and Turner *et al.* (1994)

Fire severity	Description
Unburned	Plant parts green and unaltered, no direct effect from heat
Scorched	Unburned but plants exhibit leaf loss from radiated heat
Light	Canopy trees with green needles although stems scorched Surface litter, mosses, and herbs charred or consumed Soil organic layer largely intact and charring limited to a few mm depth
Moderate or severe surface burn	Trees with some canopy cover killed, but needles not consumed All understorey plants charred or consumed Fine dead twigs on soil surface consumed and logs charred Pre-fire soil organic layer largely consumed
Deep burning or crown fire	Canopy trees killed and needles consumed Surface litter of all sizes and soil organic layer largely consumed White ash deposition and charred organic matter to several cm depth

FIRE EFFECTS on soils

Table 1 Soil properties modifiable by fires

Physical, physico-chemical, and mineralogical properties

Water repellence
Structure stability
Bulk density
pH
Particle-size distribution
Mineralogical assemblage
Colour
Temperature regime

Chemical properties

Quantity of organic matter
Quality of organic matter
Availability of nutrients
Exchangeable capacity
Base saturation

Biological properties

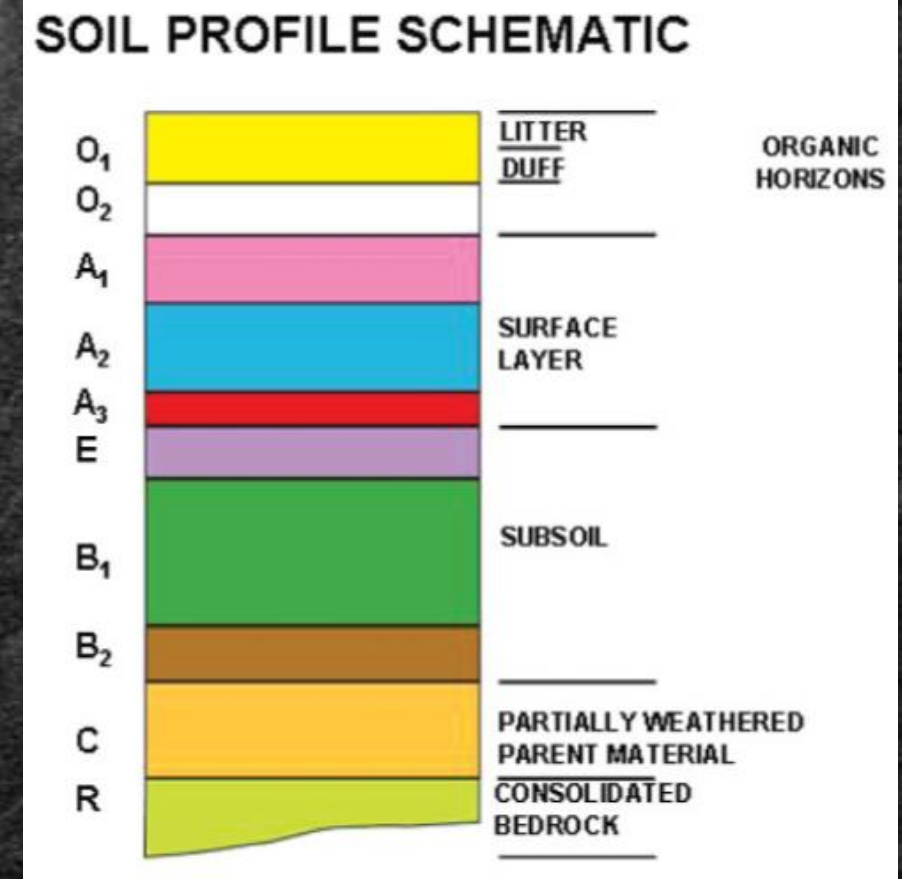
Microbial biomass
Composition of microbial community
Soil-dwelling invertebrates biomass
Composition of soil dwelling invertebrates community

FIRE EFFECTS on soils

The **effects of fire on soil properties** must be evaluated within the complex organic and inorganic matrix of the **soil profile**.

The **magnitude of change** occurring during a fire depends upon the level of **fire severity**, **combustion and heat transfer**, **magnitude and depth of soil heating**, **proximity of the soil property to the soil surface**, and the **threshold temperatures** at which the different soil properties change.

The **threshold temperature** is the temperature at which nutrients are volatilized, or that irreversible damage occurs to a particular soil property.



FIRE EFFECTS on soils

Temperature thresholds have been classified into three general classes, namely:

Relatively insensitive soil properties - do not change until temperatures have reached ca. **450 °C**).

- clays
- cations (calcium, magnesium, potassium)
- other minerals such as manganese.

Moderately sensitive soil properties – are changed at temperatures **between 100° and 400°C**.

- Sulfur
- organic matter
- soil properties dependent upon organic matter.

Sensitive soil properties – are changed at temperatures **less than 100 °C**.

- living microorganisms (bacteria, fungi, mycorrhizae)
- plant roots, and seeds
- Biologically mediated nutrient cycling processes in soils.

FIRE EFFECTS on soils

The **physical properties of soil affected by fire** include:

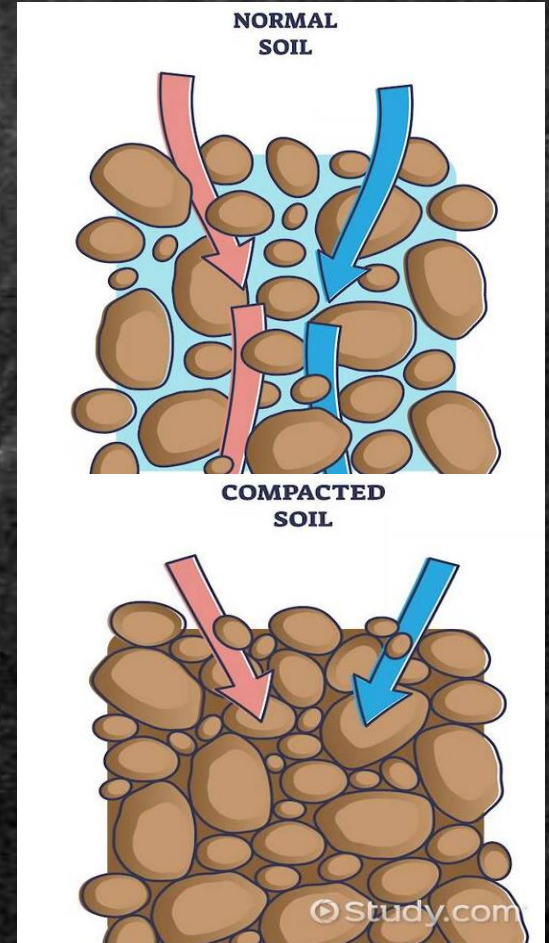
- structure
- total porosity
- pore size distribution
- water retention capacity

Soil structure results from the aggregation of mineral particles through the action of organic matter (OM). In the **upper soil horizon (A)** **structure is strongly determined by OM**.

In the **B horizon**, **structure depends more on clay minerals** and the **composition of the cation exchange complex**. The **aggregation** of particles **increases soil porosity and structure**.

Clay minerals are not significantly affected by fire because:

- they tolerate fairly high temperatures without suffering irreversible damage;
- the content of clay minerals in the top centimeters of soil is low (<5%).



FIRE EFFECTS on soils

Fire **affects chemical properties and processes** of the soil, such as:

- organic matter (OM)
- cation exchange capacity
- buffering capacity
- pH
- nutrient storage and recycling

Response of OM to temperature:

- | | |
|------------|---|
| < 100 °C | — loss of OM |
| 100–200 °C | — loss of volatile compounds |
| 200–300 °C | — loss of about 85% of soil OM due to destructive distillation |
| > 300 °C | — OM remains as carbon residues, which are lost after ignition |
| 450–500 °C | — heating the soil to 450°C for 2 hrs or 500°C for ½ hour eliminates 99% of OM. |

FIRE EFFECTS on soils

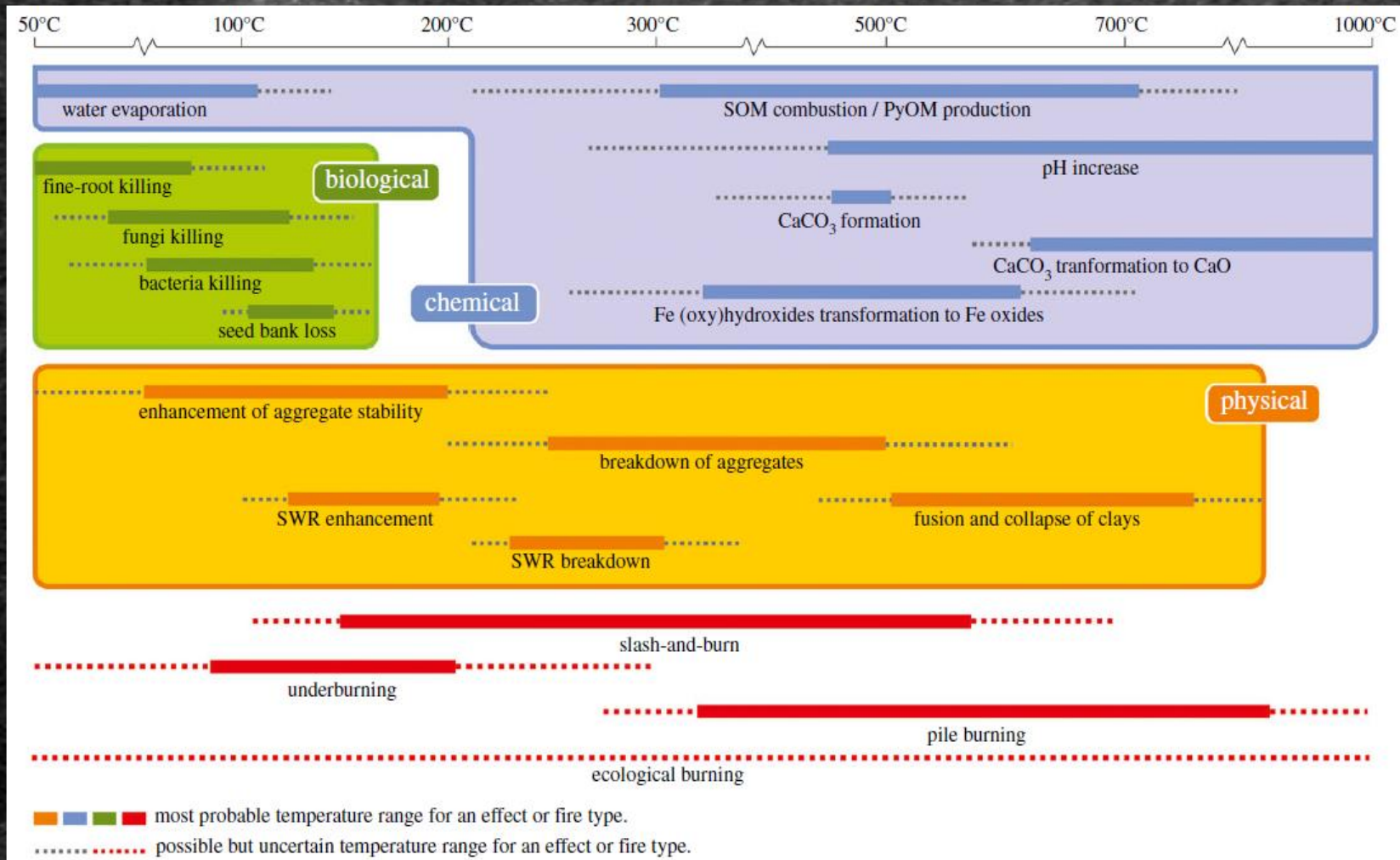


Figure 1. Effects on the biological, chemical and physical properties of soil and associated temperature ranges reached near the mineral soil surface for different types of **human-induced fires (slash-and-burn, underburning, pile burning and ecological burning)**.

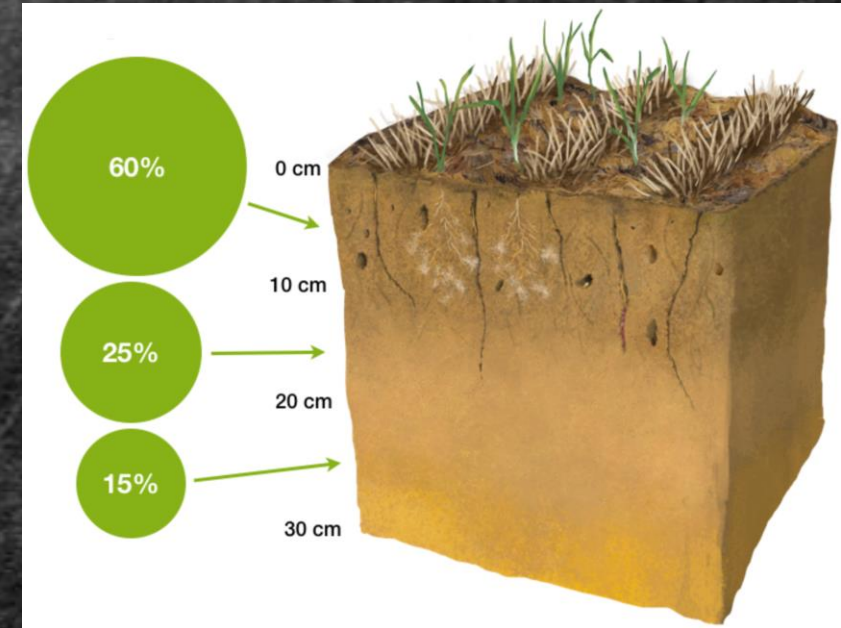
The temperature ranges provided are broad estimates. Specific temperatures and associated soil effects will depend on the characteristics of each fire and soil.

Temperature scale is nonlinear.
SOM: soil organic matter; PyOM: pyrogenic organic matter; SWR: Soil water repellency.

FIRE EFFECTS on soils

Fire effects on the organic matter (OM) component of soil structure **are more severe** because:

- the destruction of OM begins at low temperatures (200 °C)
- OM is concentrated in the surface layers of the soil profile, where it is more exposed to heat.



The collapse of soil structure reduces its total porosity. The **loss of macropores in the A horizon hinders water infiltration** and **increases surface runoff**. Water is retained by capillarity, and the retention capacity is greater the higher the volume of micropores. Changes in porosity also impair the soil's ability to retain water.

The **heating of OM at the soil surface** can cause strong **hydrophobicity** (water repellency), which **hinders water infiltration** in some burned areas.

FIRE EFFECTS on soils

Fire alters the nutrient cycling processes in ecosystems by replacing long-term biological decomposition rates with those of **instantaneous thermal decomposition (pyrolysis)** that occurs during the combustion of organic fuels.

Many nutrients essential for plant growth including N, P, S, and some cations are affected to some extent by fire. Nitrogen is likely the most limiting nutrient in natural systems, followed by P and S.

Cations released by burning may affect soil pH and result in the immobilization of P.

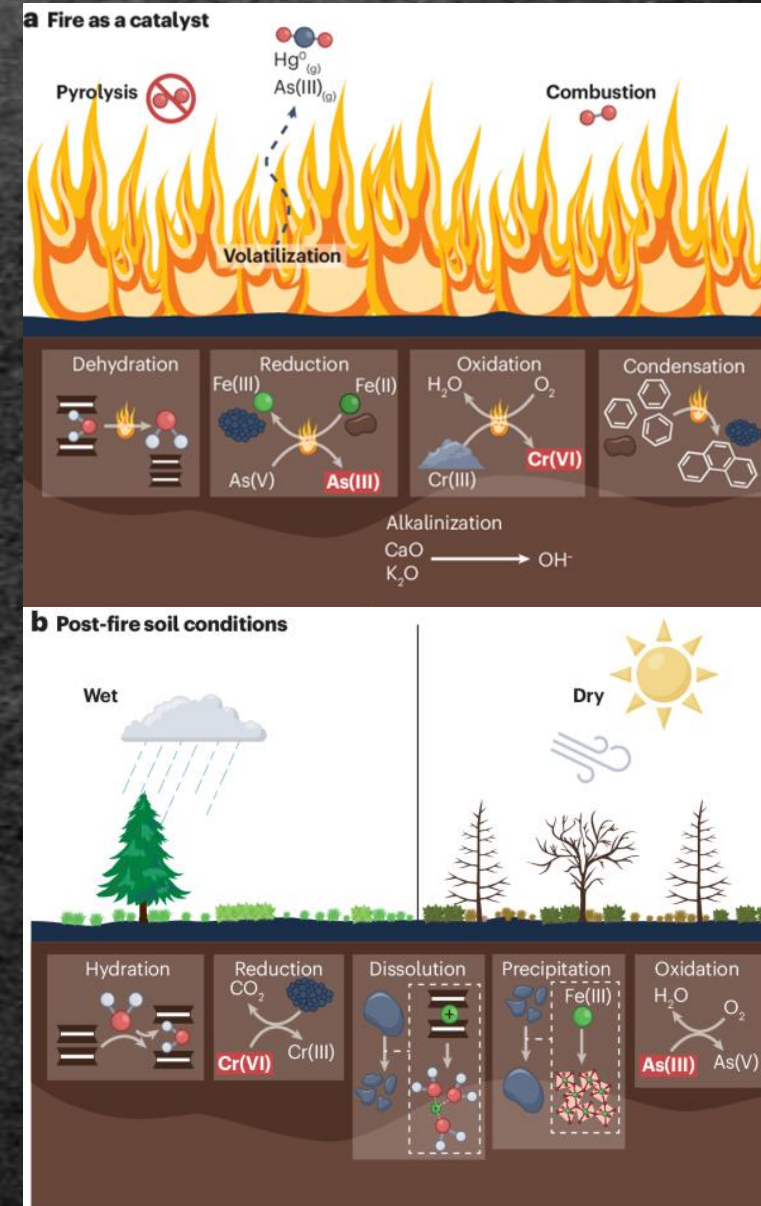


Fig. 5: Wildfire-induced chemical changes and their impacts.

FIRE EFFECTS on soils

The **main mechanisms of nutrient loss** are:

- Direct volatilization into the atmosphere during the fire. Nitrogen is transformed into N_2 and other nitrogen gases.
- Loss of particulate material in smoke. Affects phosphorus and cations.
- Deposition of ash on the soil surface. Nutrients become highly available but may be leached into the soil or dispersed by wind.
- Surface runoff and erosion.
- Leaching into and through the soil profile.
- *In situ* retention in partially burned vegetation.

Nutrients like nitrogen are easily volatilized and lost during fire, while **others** (Ca, Mg, and Na) **remain in the ash** in oxidized forms.

FIRE EFFECTS on soils

The **soil nutrient content** in **Mediterranean drylands affected by wildfires varies** with:

- vegetation type,
- period of time after fire,
- fire recurrence and fire intensity.

Immediately after a fire, the **nutrient content** in both the **O and the A horizon** usually **increases** because of ash deposition, nutrient mineralization and formation of stable forms such as black carbon, charcoal and apatite.

Ash deposition and persistence on the soil surface is one of the most important factors in determining the soil nutrient content immediately after a fire as well as in the short- and long-term.

In the long-term, the **soil nutrient content usually decreases** because of nutrient **volatilization**, **transformation of the nutrients to more recalcitrant forms**, and **ash removal** by wind and surface runoff.

FIRE EFFECTS on soils

The **quality and quantity of the ash depend on vegetation type and fire severity**. Higher fire severity, assessed by lighter ash color, can lead to different chemical compositions of the ash.

Depending on **fire intensity**, P can be converted to apatite $\text{Ca}_5(\text{PO}_4)_3(\text{F}, \text{Cl}, \text{OH})$ which is **not plant-available** in the short term. C can be converted to black C (BC) and charcoal, **recalcitrant forms of C resistant to decomposition**.

N and C are often lost by volatilization at temperatures **between 200–500 °C**. **Ca** and **Mg** are **not lost by volatilization** due to their high volatilization temperatures.

High-intensity fires can lead to the loss of almost 80% of the total exchangeable cations in the soil through surface runoff and subsurface flow.

The depth of burning in the soil, influenced by fire intensity, affects soil organic C (SOC) changes in the A horizon. Carbon combustion starts at 200 °C.

FIRE EFFECTS on soils

Frequent fires can hinder vegetation recovery and foster land degradation and desertification. Various studies have shown that Mediterranean-type shrublands affected by three wildfires have significantly lower SOC and N content than those affected by only one fire.

The NO_3^- – N content in the O horizon can increase after a fire in shrublands, but different trends are observed with recurrence depending on the time period between fires and litter accumulation. **A 3-yr interval can lead to a decrease after the second fire, while a 9-yr interval can show an increase after the second fire**.

The NH_4^+ – N content in soil under shrubs **decreases with increasing fire recurrence**, with plots affected by three wildfires showing significantly lower contents.

The available P content in the A horizon of shrublands tends to **decrease with increasing fire recurrence**, with plots having only one fire showing higher P contents than those with two or three.

FIRE EFFECTS on soils

Table 1

Change in soil nutrient content of burnt plots vs. unburnt plots immediately after a fire. Symbols refer to: (↓) decrease, (↑) increase, (↔) no significant change, (n.d.) no data available.

	Pine and oak forests		Shrublands and grass patches	
	O horizon	A horizon	O horizon	A horizon
SOC	↓	n.d.	↓	↔
Total N	↓	↓	↓	↓
NH ₄ ⁺ -N	↑	↑	n.d.	↑
NO ₃ ⁻ -N	n.d.	n.d.	n.d.	↑
Total P	↓	n.d.	n.d.	n.d.
Available P	n.d.	↑	↓	↓
Exchangeable Ca ²⁺	↓	↓	n.d.	↑
Exchangeable Mg ²⁺	↓	↓	n.d.	↑
Exchangeable Na ⁺	n.d.	↓	n.d.	↑
Exchangeable K ⁺	↓	↓	n.d.	↑

Table 2

Change in soil nutrient content of burnt plots vs. unburnt plots in the short-term (samples collected within one year after a fire). Symbols refer to: (↓) decrease, (↑) increase, (↔) no significant change, (n.d.) no data available.

	Pine and oak forests		Shrublands and grass patches	
	O horizon	A horizon	O horizon	A horizon
SOC	n.d.	↓	n.d.	↓
Total N	↓	↓	n.d.	↓
NH ₄ ⁺ -N	n.d.	↓	n.d.	↑
NO ₃ ⁻ -N	↑	↑	n.d.	↓
Total P	↓	↑	↓	↓
Available P	n.d.	↑	n.d.	↑
Exchangeable Ca ²⁺	↓	↑	n.d.	↓
Exchangeable Mg ²⁺	↓	↑	n.d.	↔
Exchangeable Na ⁺	n.d.	↑	n.d.	↔
Exchangeable K ⁺	↓	↑	n.d.	↑

Table 3

Change in soil nutrient content of burnt plots vs. unburnt plots in the long-term (samples collected later than one year after a fire). Symbols refer to: (↓) decrease, (↑) increase, (↔) no significant change, (n.d.) no data available.

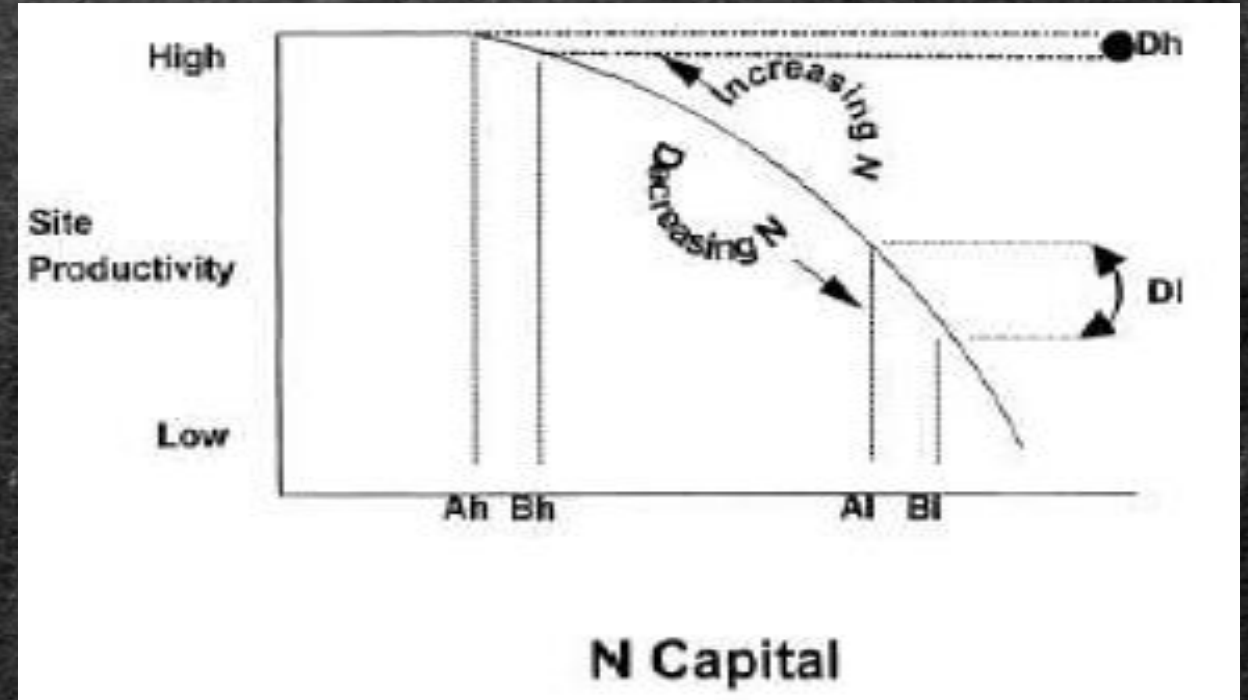
	Pine and oak forests		Shrublands and grass patches	
	O horizon	A horizon	O horizon	A horizon
SOC	n.d.	↓	↑	↓
Total N	n.d.	↓	↑	n.d.
NH ₄ ⁺ -N	n.d.	n.d.	n.d.	n.d.
NO ₃ ⁻ -N	n.d.	n.d.	n.d.	n.d.
Total P	n.d.	↑	↑	↑
Available P	n.d.	↑	n.d.	↓
Exchangeable Ca ²⁺	n.d.	n.d.	↑	n.d.
Exchangeable Mg ²⁺	n.d.	n.d.	↑	↓
Exchangeable Na ⁺	n.d.	n.d.	n.d.	n.d.
Exchangeable K ⁺	n.d.	n.d.	↑	↓

SOC – soil organic carbon

Changes in soil nutrients with time after fire in Mediterranean forests, shrublands, and grasslands.

FIRE EFFECTS on soils

The **importance of nutrient changes caused by fire is directly linked to the ecosystem's net primary productivity (NPP)**. For example, in a low-to-moderate intensity fire, a given change in the nitrogen capital of a highly productive system (from Ah to Bh) does not significantly affect post-fire productivity (Dh).

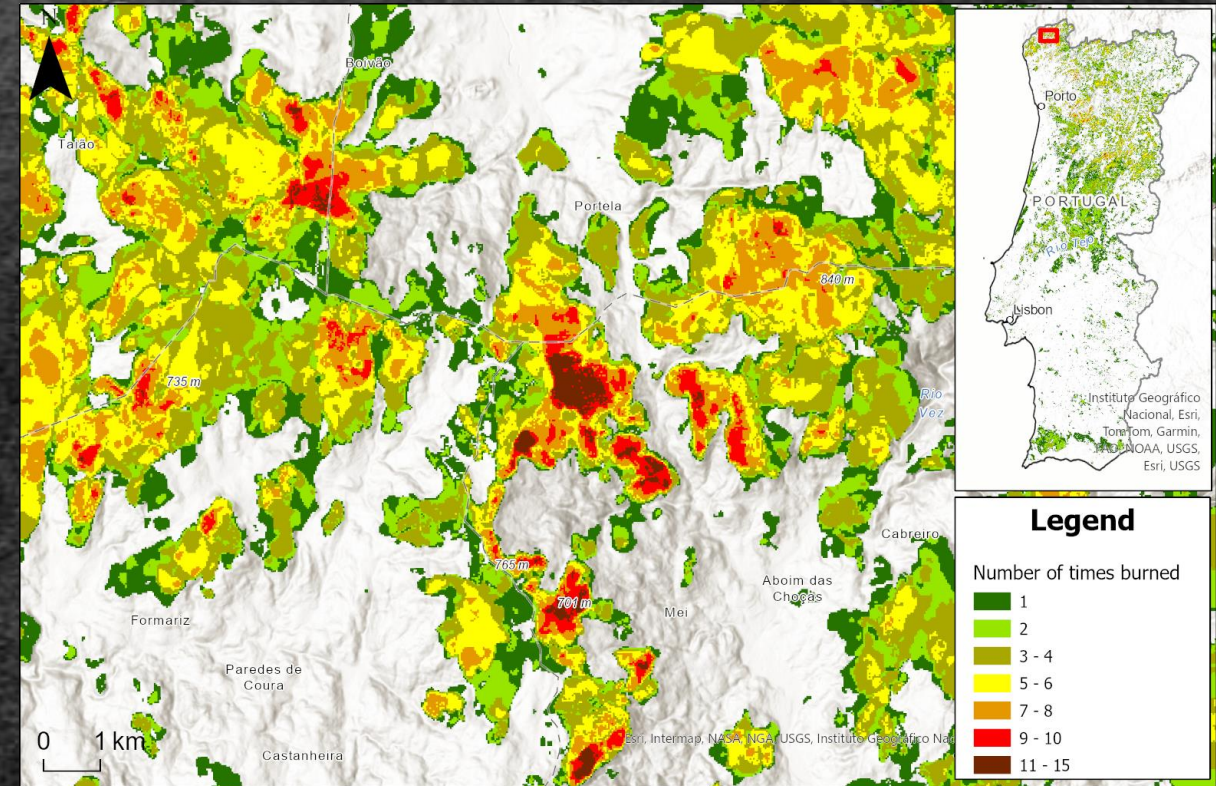


A similar change in nitrogen capital (from Al to Bl) in a nutrient-poor system causes a much greater loss in productivity (DI). Therefore, **nitrogen richness and ecosystem productivity can be used as predictive indicators of an ecosystem's potential for recovery after fire.**

FIRE EFFECTS on soils

In conclusion, wildfires in Mediterranean ecosystems have **complex impacts on soil nutrient levels** that **vary significantly based on fire characteristics, the specific nutrient, the soil horizon, vegetation type, and the timescale considered.**

While an **initial increase in nutrients can occur due to ash deposition, longer-term effects often involve nutrient losses** and alterations in nutrient availability, especially with high fire intensity and frequency.



1984 - 2023

FIRE EFFECTS on soils

Table 2 Summary of the effects of fires on properties of forest soils

PHYSICAL, PHYSICO-CHEMICAL, AND MINERALOGICAL PROPERTIES

Water repellence: the natural water repellence of soil often increases because of the formation of a continuous water-repellent layer a few cm beneath the surface. It implies limitations in soil permeability and, thus, increased runoff and erosion

Structure stability: complexity decreases as a result of the combustion of organic cements

Bulk density: increases because of the collapse of aggregates and the clogging of voids by the ash and the dispersed clay minerals; as a consequence, soil porosity and permeability decrease

Particle-size distribution: does not change directly, but the increased erosion can remove selectively the fine fraction

pH: in noncalcareous soils increases, although ephemerally, because of the release of the alkaline cations (Ca, Mg, K, Na) bound to the organic matter

Mineralogical assemblage: changes, but only at temperatures higher than 500C

Colour: darkens, due to charring, and reddens, due to formation of iron oxides

Temperature regime: changes temporarily because of both the disappearing of the vegetable mantle and the darkening of ground (decreased albedo)

CHEMICAL PROPERTIES

Quantity of organic matter: decreases immediately after fire, but in the long run generally exceeds the pre-fire level

Quality of organic matter: changes remarkably, with a relative enrichment of the fraction more recalcitrant to biochemical attack. This is due to both selective burning of fresh residues (leaves, twigs, etc.), and neoformation of aromatic and highly polymerised (humic-like) compounds. Charred material, an exclusive product of incomplete combustion, shows residence times of centuries or even millennia

Availability of nutrients: increases, often remarkably, but ephemerally

- Organic N (unavailable, often almost coinciding with total N) in part volatilises and in part mineralises to ammonium, a form available to biota. Ammonium adsorbs on negative charges of mineral and organic surfaces but, with time, is destined to be biochemically transformed to nitrate, which is leached soon if not taken up by cells. Nitrogen availability lowers to pre-fire levels in a few years
- Organic P mineralises to orthophosphate and the loss through volatilisation is negligible; orthophosphate is not leached out of soil but if not promptly taken up, it precipitates as slightly available mineral forms
- Calcium, magnesium and potassium often increase remarkably but ephemerally

Exchange capacity: decreases proportionally to the loss of organic matter

Base saturation: increases as a consequence of the prevailing release of bases from the combusting organic matter

BIOLOGICAL PROPERTIES

Microbial biomass: decreases; the recovery of the pre-fire level depends chiefly on promptness of plant recolonisation

Composition of microbial community: changes as a consequence of the selective effect of fire on some groups of microorganisms and the modification imposed to vegetation; generally, fungi diminish more than bacteria

Soil-dwelling invertebrates biomass: decreases, but less than that of microorganisms thanks to the higher mobility of the invertebrates

Composition of soil-dwelling invertebrates community: changes, the time of recovery of the pre-fire assemblage differs highly among the various phyla

Certini, G. (2005). Effects of fire on properties of forest soils: a review. *Oecologia*, 143, 1-10.

FIRE EFFECTS on soils

Incluir hidrofobicidade

E era outra coisa de que não me lembro...

FIRE EFFECTS on water

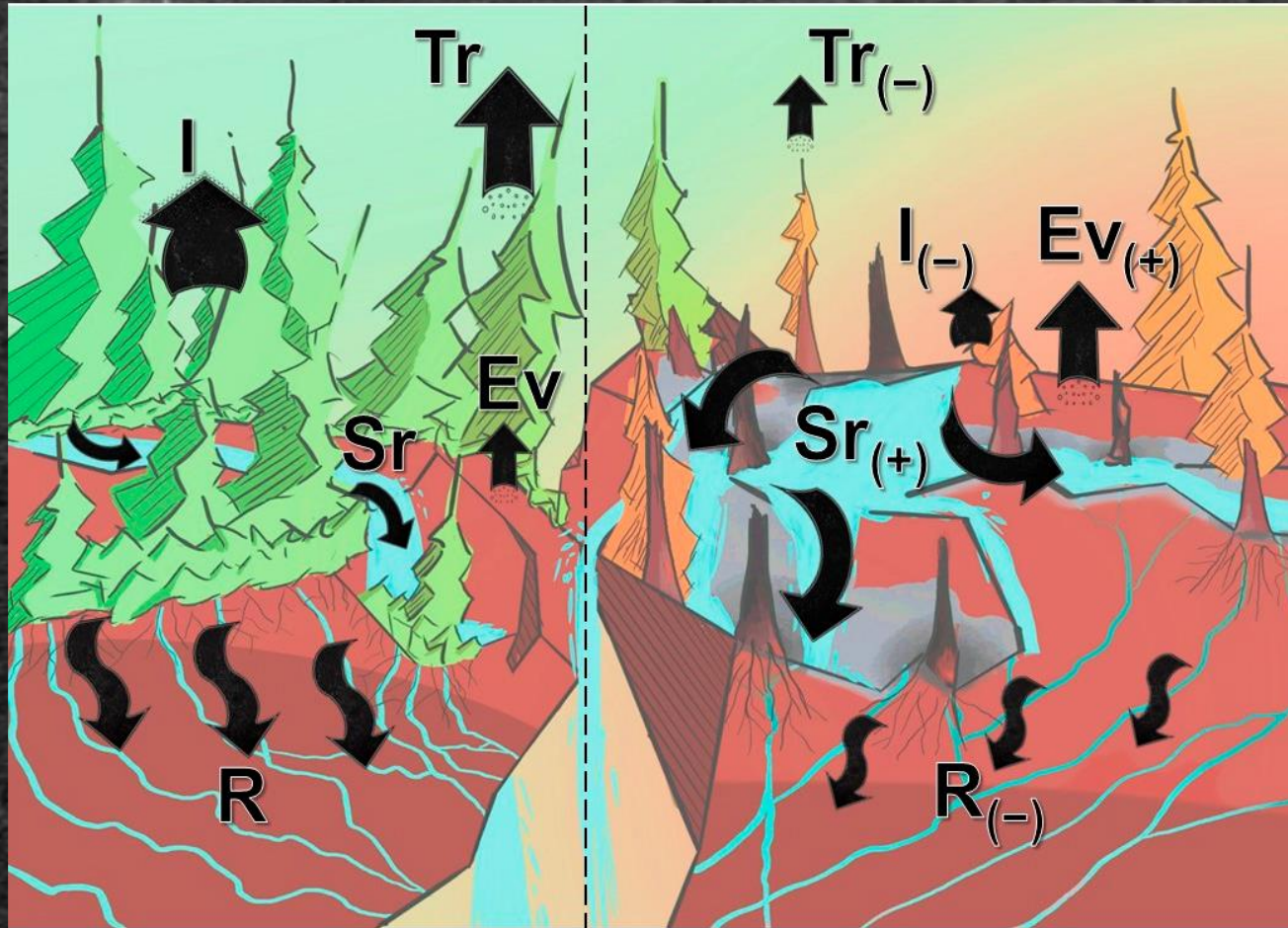


Figure 1. Schematic representation of **hydrological processes in a forest ecosystem, illustrating the hypothesis that altered water-balance components contribute to reduced groundwater recharge after a wildfire**. Subfigure (a) shows the region before the wildfire, highlighting normal hydrological processes. Subfigure (b) focuses on the same after being heavily affected by high-intensity wildfires, detailing the cascading effects on water cycle components: interception (I), transpiration (Tr), evaporation (Ev), surface runoff (Sr), and recharge (R).

FIRE EFFECTS on water

Rainfall Interception ↓ (Decreases)

Vegetation loss from fire (especially canopy and understory) leads to a sharp reduction in rainfall interception. Less interception means more water reaches the ground, often intensifying the hydrological response. Ash deposits on foliage (if any remains) usually do not significantly increase interception.

Infiltration ↓ (Often Decreases)

Fire can cause soil water repellency (hydrophobicity), especially in high-intensity fires, reducing the soil's capacity to absorb water. Ash may temporarily clog pores, further inhibiting infiltration. Soil structure may degrade (e.g., loss of organic matter and aggregates), reducing macroporosity. However, infiltration can increase in low-severity fires if litter is removed but soil properties remain intact.

FIRE EFFECTS on water

Surface Runoff ↑ (Increases)

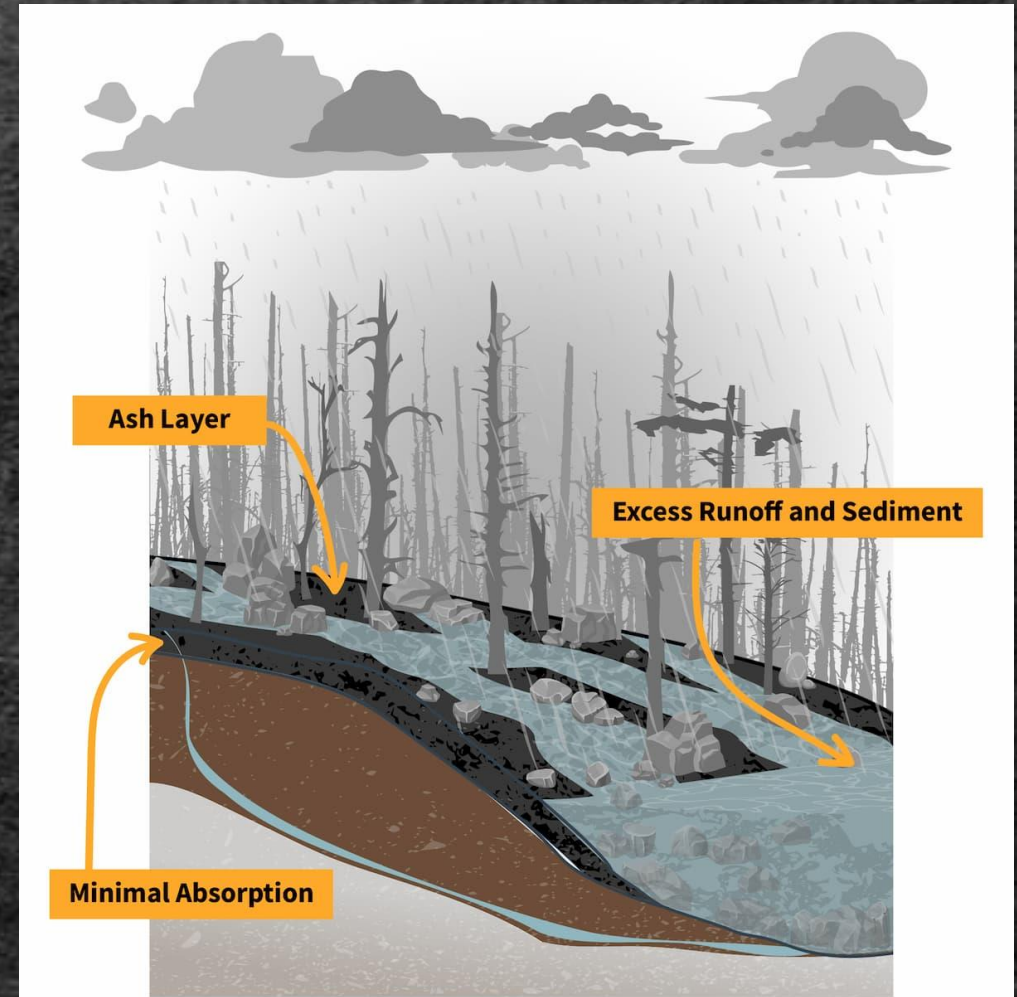
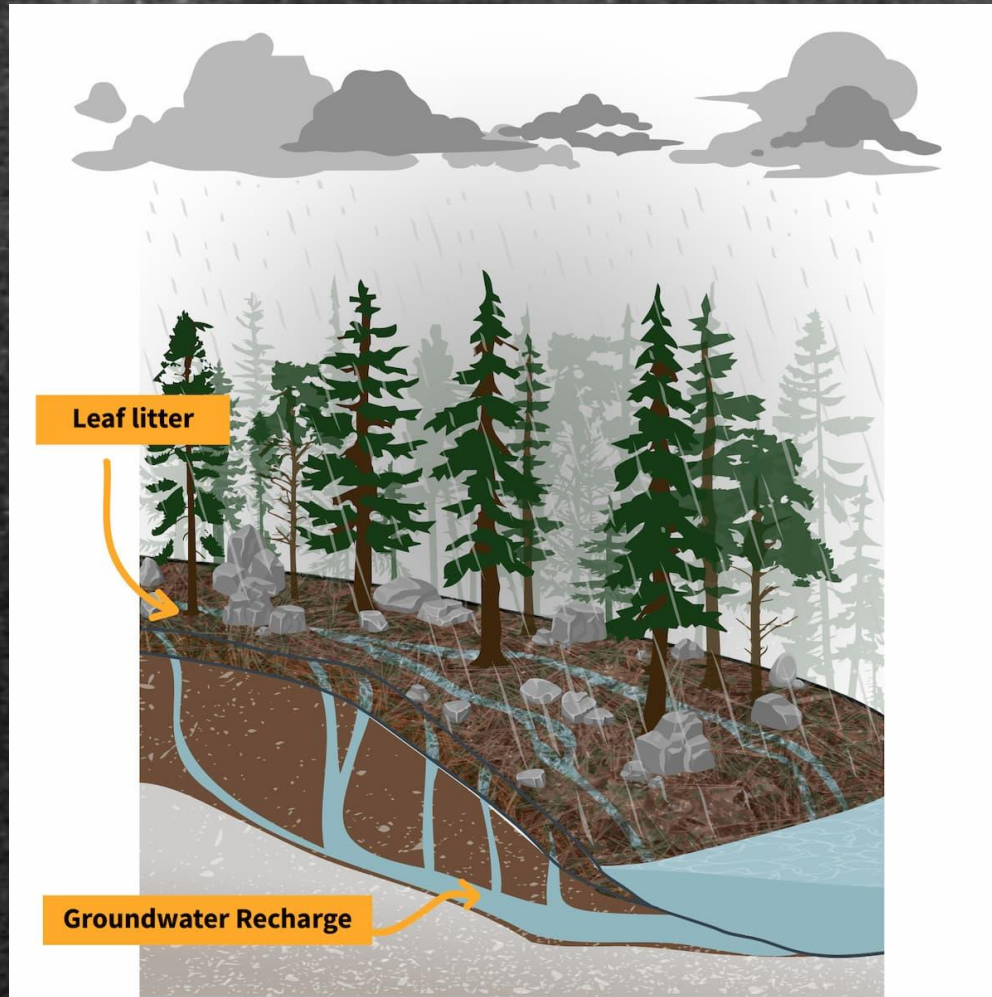
With interception and infiltration both reduced, more rainfall becomes surface runoff. Steeper slopes and intense rainfall post-fire can lead to flashier responses and higher peak flows. The result is a higher risk of erosion, sediment transport, and flooding.

Other Notable Effects

Evapotranspiration ↓ due to canopy loss and vegetation mortality. Soil erosion ↑, especially in the first 1–2 years post-fire, due to exposed soil and increased runoff.

Streamflow ↑, especially during wet seasons following fire, due to reduced evapotranspiration and enhanced runoff. Water quality ↓ with increases in sediment, nutrients (e.g., nitrogen, phosphorus), and ash entering streams.

FIRE EFFECTS on water



FIRE EFFECTS on water

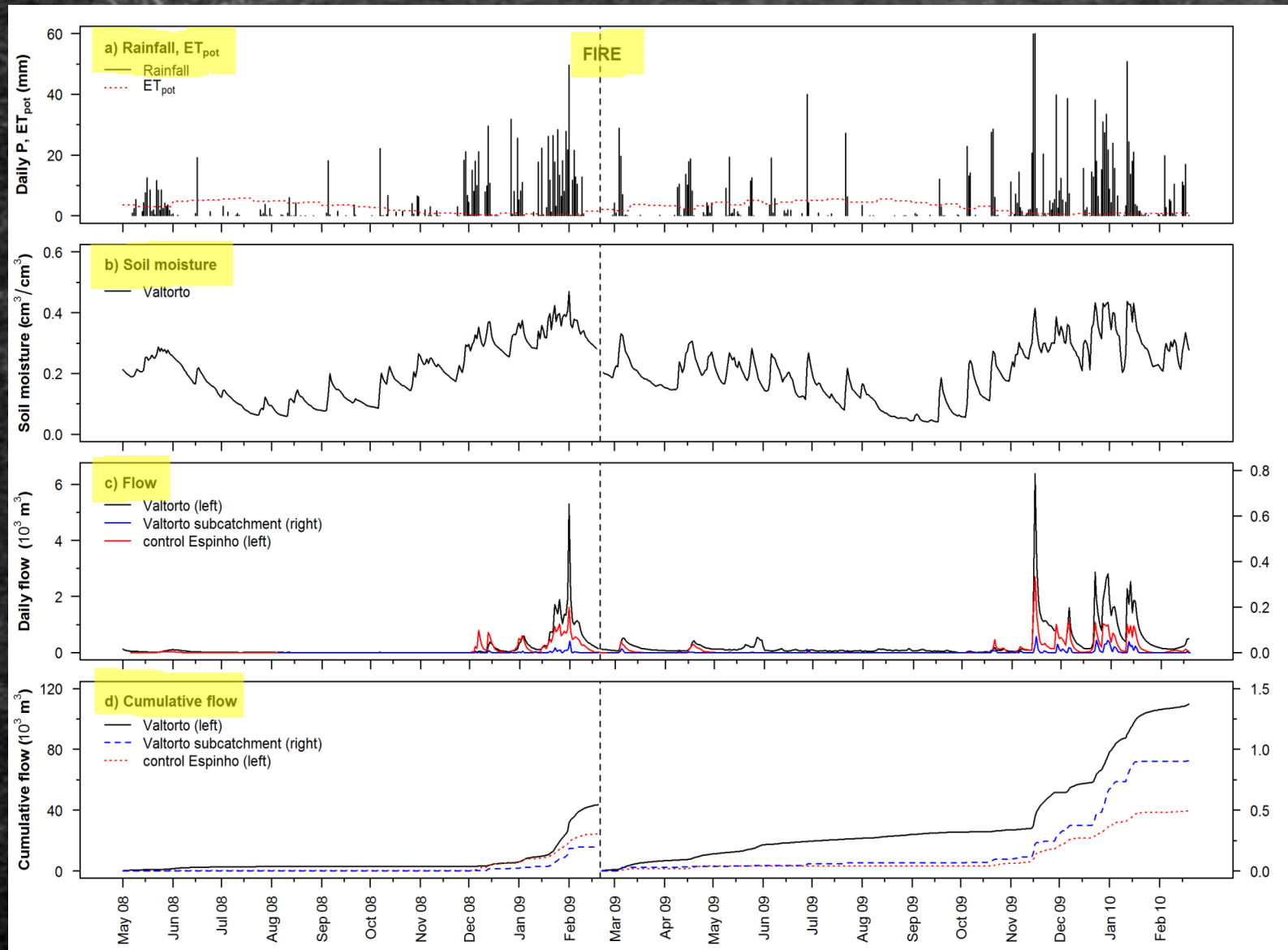


Fig. 2. Time series of (a) daily rainfall (P) and potential evapotranspiration (ET_{pot}), (b) catchment average soil moisture content, (c) streamflow, and (d) cumulative streamflow before and after the experimental fire on 20 February 2009 (vertical dashed line).

Note that **only the Valtorto catchment was burned; Espinho is the unburned control catchment**. Also note that in the streamflow graphs (c, d), the values on the primary y-axis (left) apply to the Valtorto and Espinho main catchments, while the values on the secondary y-axis (right) apply to the Valtorto subcatchment.

Increased post-fire flow and cumulative flow.

FIRE EFFECTS on water

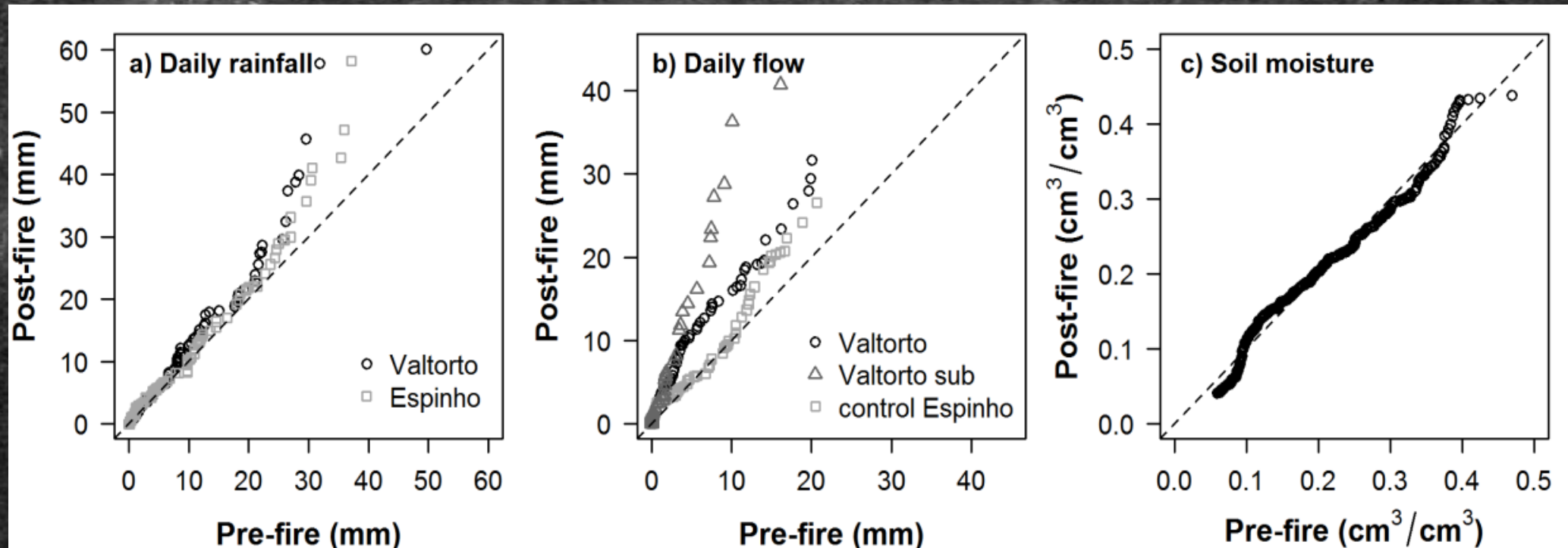


Fig. 3. QQ-plots of (a) daily rainfall, (b) streamflow and (c) soil moisture in the Valtorto (burned) and Espinho (control) catchments, comparing the quantiles of pre- and post-fire distributions relative to the $y = x$ line (dashed). Where plotted data deviate from the $y = x$ line, pre- and post-fire values are different. **The graphs show that post-fire rainfall (a) and flow (b) was higher than pre-fire for all catchments, while the soil moisture distribution (c) remained largely unchanged.** To facilitate comparison between the different catchments and scales, flow volumes in graph b) are given in mm.

FIRE EFFECTS on water

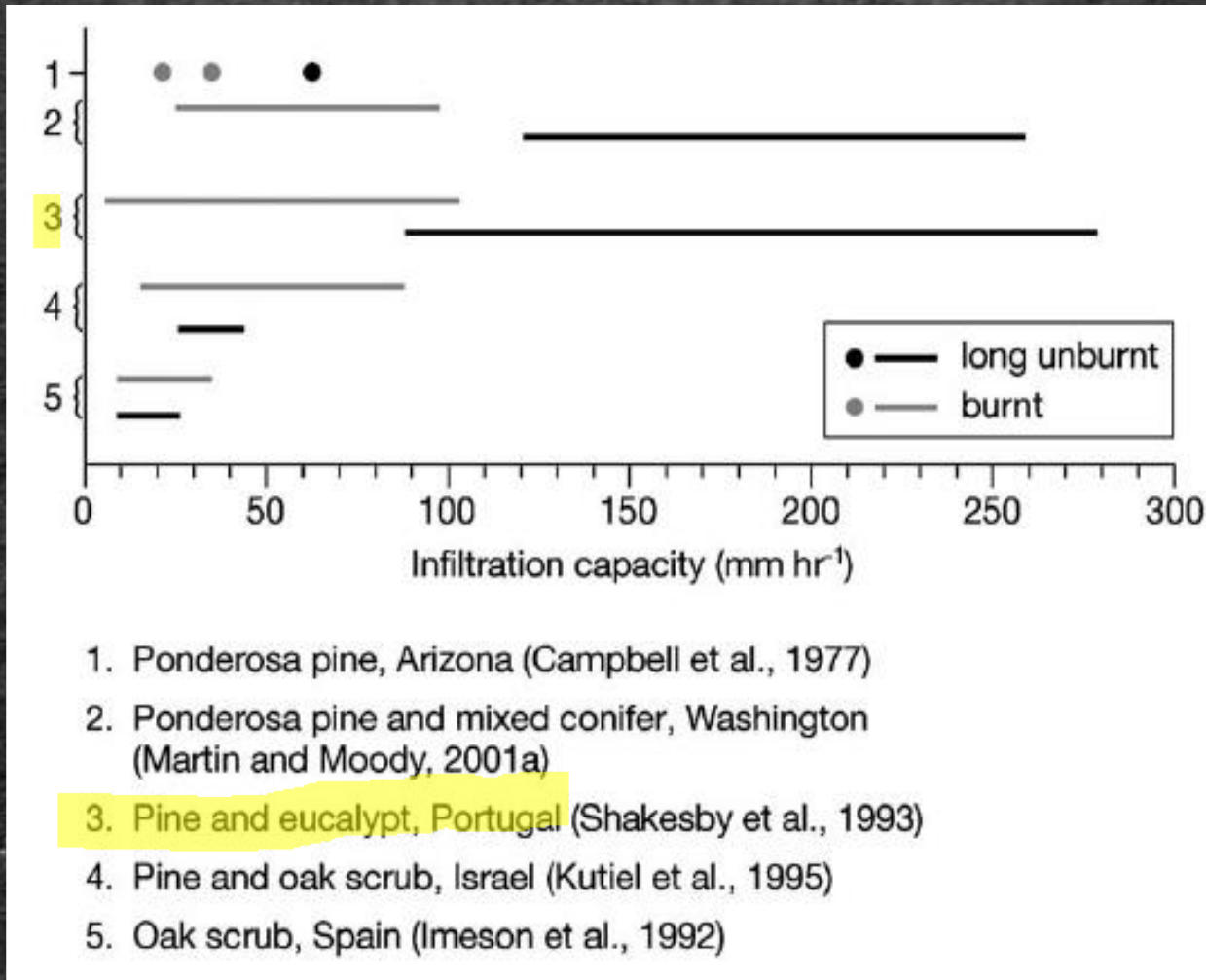


Fig. 2. **Infiltration capacities measured after wildfire and on comparable unburnt terrain** according to various authors. Lines represent ranges of values and points represent individual values.

Decreased infiltration capacity in burned pine and eucalypt forests, in Portugal.

FIRE EFFECTS on water

Higher post-fire erosion rates, increasing with fire intensity.

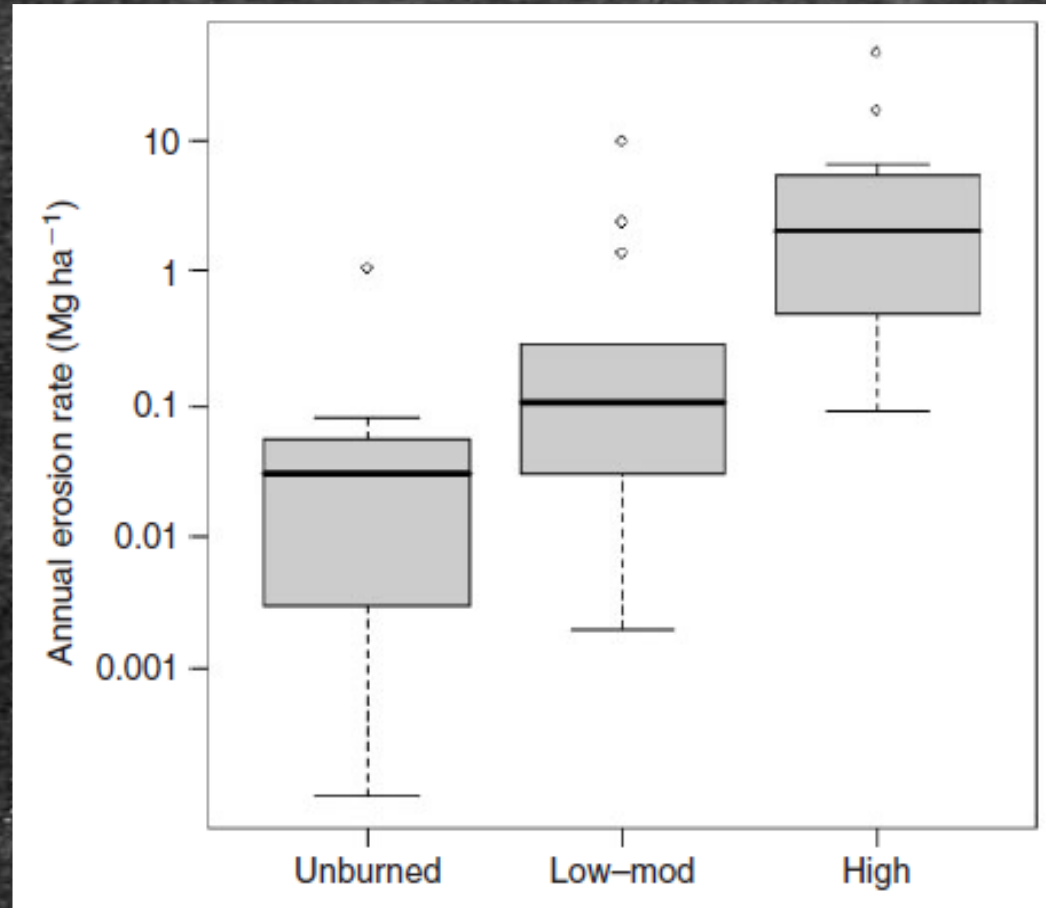
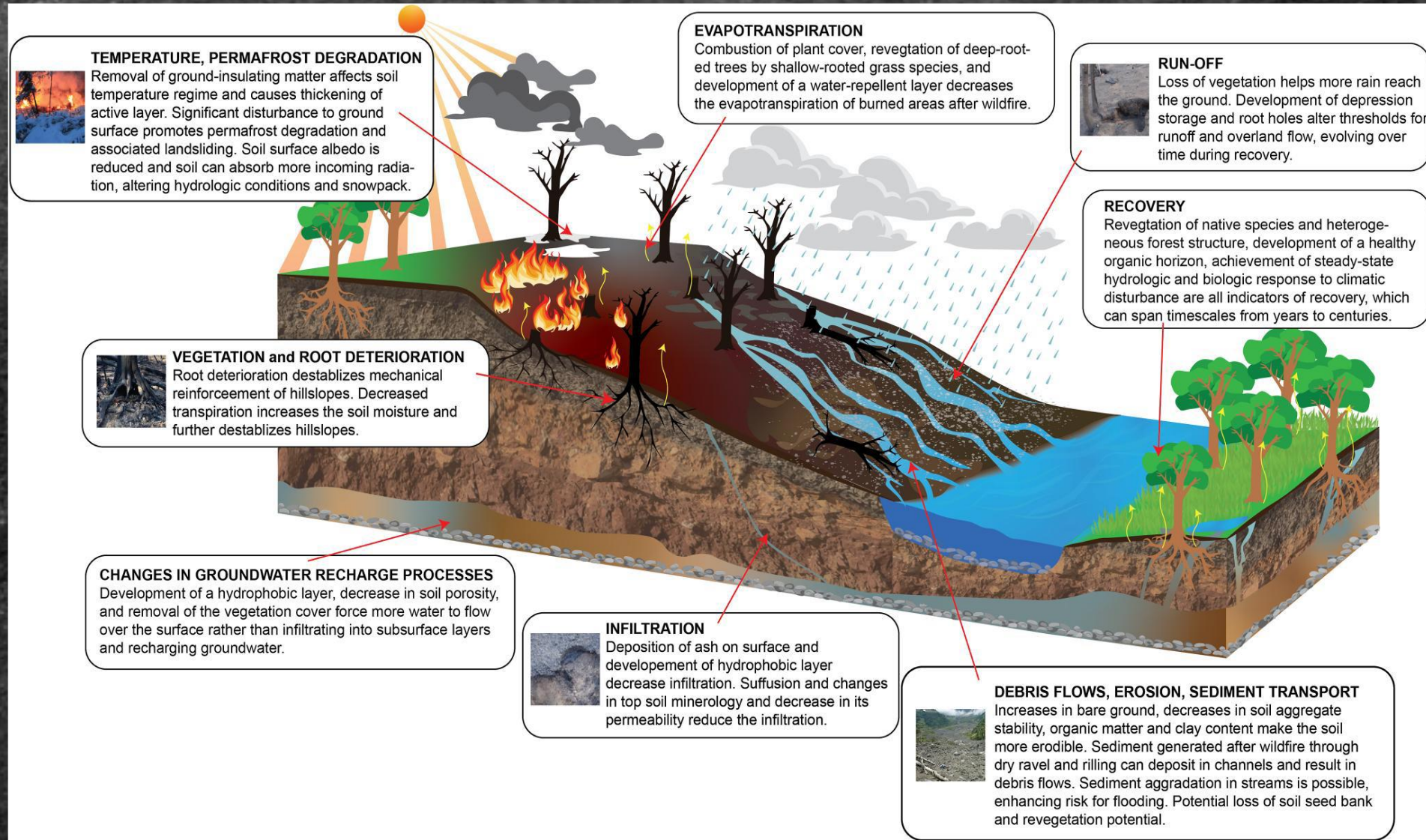


Fig. 5. Box plot statistics of **annual erosion rates (Mg ha⁻¹)** from different studies in the Mediterranean basin **aggregated by fire severity** (unburned, low-to-moderate fires, and high-severity fires). Annual erosion rates are computed from studies in Table 1 that monitored erosion for a period shorter or equal to 2 years after fire.

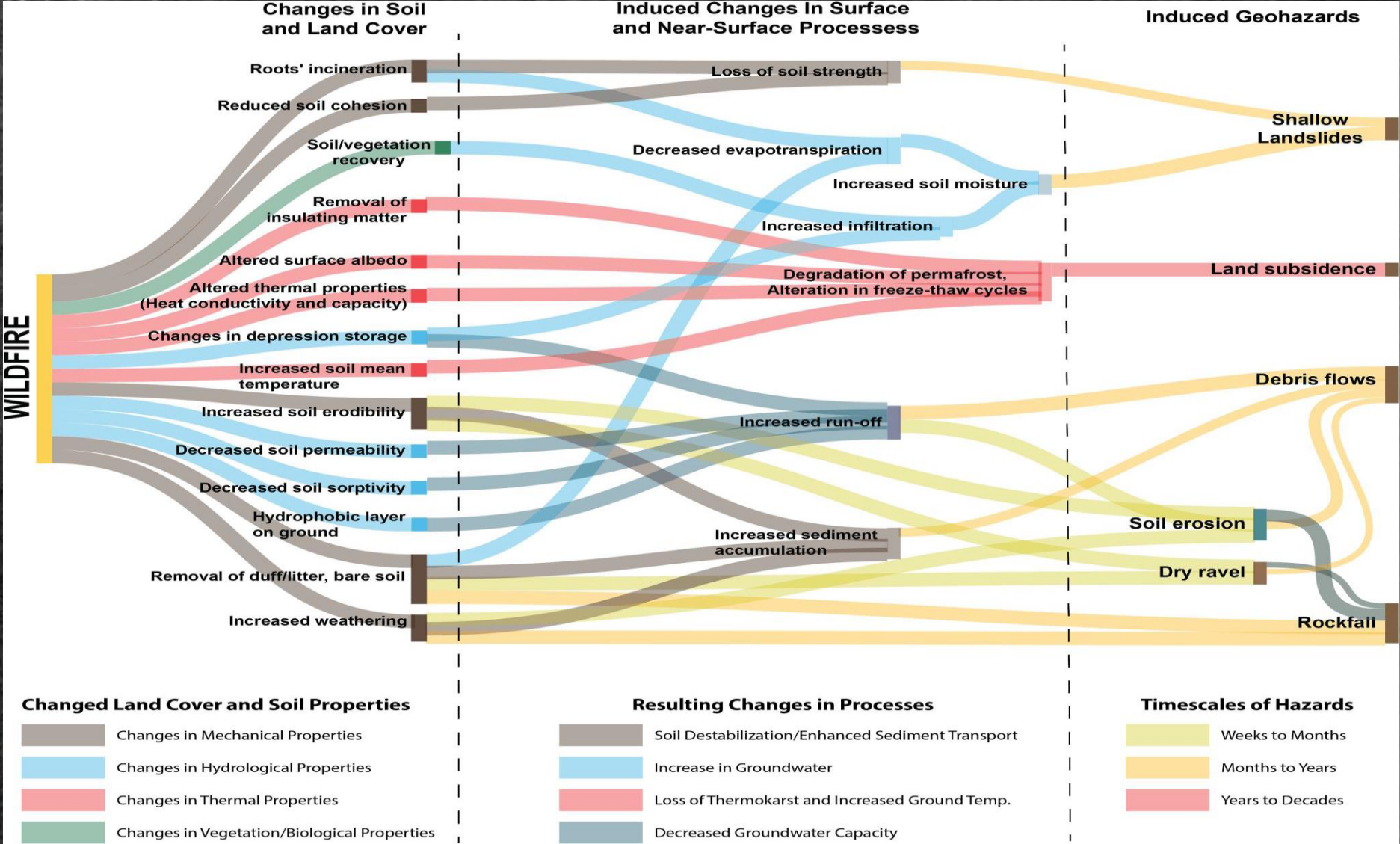
FIRE EFFECTS on water

Figure 1



FIRE EFFECTS on water

Figure 2



FIRE EFFECTS on water

Enhanced nutrient losses by runoff after fire are related to the lack of protective soil cover, resulting from the partial or complete consumption of vegetation and litter layer.

The **decrease in interception by vegetation and litter** not only enhances the erosive potential of rain but also increases the effective rainfall intensity, **increasing the likelihood of overland flow (surface runoff)** generation and the associated **transport of ash, charcoal and soil particles as well as of nutrients in dissolved and particulate form**.

Aside from decreasing the on-site nutrient pools, runoff-induced nutrient losses **enhance the risk of eutrophication to downstream water bodies** and the associated **deterioration of chemical and biological water quality**.

FIRE EFFECTS on water

Table 10
Post-fire nutrient losses expressed as percentages of soil nutrient reserves^a

	Falgorosa burnt eucalyptus Mean soil depth		Lourizela burnt pine Mean soil depth	
	3 cm	10 cm	23 cm	50 cm
Total nitrogen	6%	2%	0.6%	0.4%
Available phosphorus ^b	55%	19%	6.5%	3.7%
Exchangeable K	11%	3.7%	0.7%	0.5%

^aLosses over both years of monitoring (1st and 2nd years after fire at Falgorosa and 2nd and 3rd years after fire at Lourizela)—average of both plots adjusted to mean annual rainfall at each site.

^bFirst year of data collection only.

Fires in eucalyptus and pine forests in 1992 and 1993, respectively, in the Águeda Basin, northern Portugal, **increased eroded nutrient losses in sediment by 3–4 orders of magnitude. The enhanced rates of loss were sustained for at least 3 years, after which data from the burnt pine site and from plots previously burned in 1986 suggest that losses begin to decline.** N losses were largest followed by much smaller losses of exchangeable K and available P. However, **losses of available P at both sites constituted the biggest proportion of soil nutrients and are likely to have the greatest adverse affect on soil fertility.**

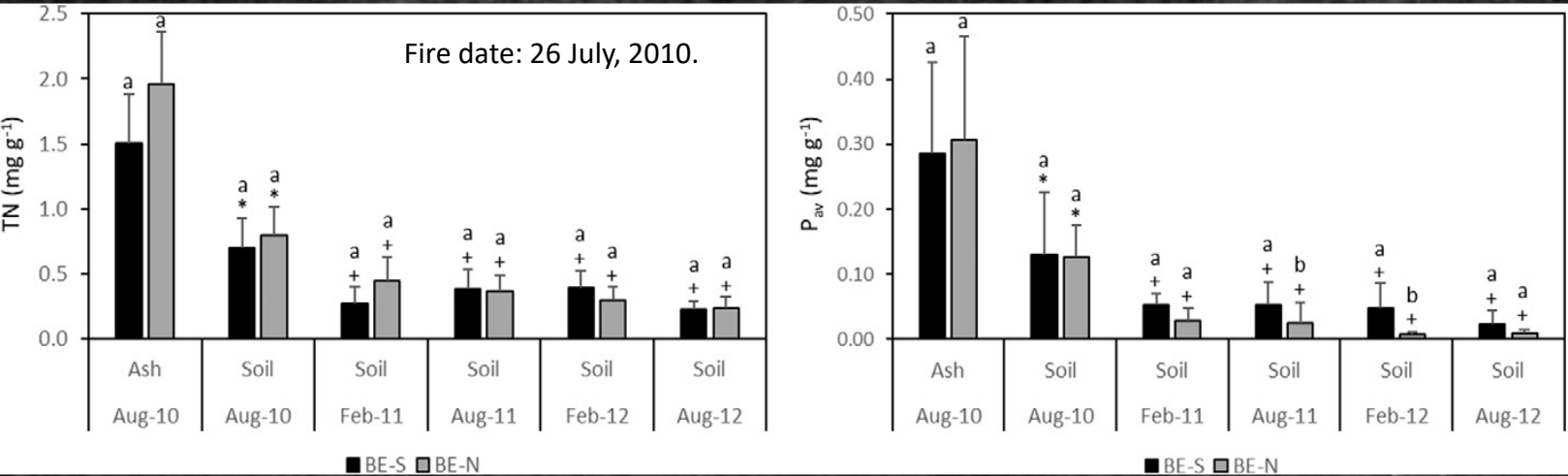


Fig. 5. Average ($\pm$ standard deviation; $n = 5$) total nitrogen (TN) and available phosphorus (P_{av}) concentrations (mg g⁻¹) in the ash and topsoil (0–2 cm) layers of the two study sites during the first two years after fire. BE-S — burnt eucalypt site facing south-east and BE-N — burnt eucalypt site facing north-west. Different letters correspond to significant differences ($p=0.05$) between sites. Different symbols correspond to significant differences ($p=0.05$) between sampling events.

Thomas, A. D., Walsh, R. P., & Shakesby, R. A. (1999). Nutrient losses in eroded sediment after fire in eucalyptus and pine forests in the wet Mediterranean environment of northern Portugal. *Catena*, 36(4), 283-302.

Serpa, D., Ferreira, R. V., Machado, A. I., Cerqueira, M. A., & Keizer, J. J. (2020). Mid-term post-fire losses of nitrogen and phosphorus by overland flow in two contrasting eucalypt stands in north-central Portugal. *Science of the Total Environment*, 705, 135843.

FIRE EFFECTS on water

The increase in nutrient loss is attributed to increased erosion as a result of bare areas resulting from combustion of the litter layer and ground vegetation, and the easily erodible surface layer of non-cohesive ash containing high concentrations of nutrients.

The nutrient losses from the eucalyptus and pine forests of the Águeda Basin suggest that **recovery to pre-fire nutrient levels would take up to five years, depending on the rate of revegetation.**

This is considerably **longer than recovery times reported elsewhere in the drier areas of the Mediterranean**, probably because **the effect of post-fire vegetation recovery is more than outweighed by the greater erosivity associated with a wet Mediterranean environment**, at least with the land management of the area.

FIRE EFFECTS on plants

Most **plant cells die** if heated to **$50^{\circ} < T < 55^{\circ}\text{C}$** . Plant tissue withstands heat in a time-T dependent manner. Mortality can occur at high T after a short period, while death at lower T requires a longer exposure.

Some plant tissues, particularly **meristems or buds** are much **more sensitive to heat when actively growing** and their **tissue moisture is high**, than when their moisture content is low.

Plant mortality depends on the amount of meristematic tissues killed. Susceptible tissue may not be exposed to heating by fire because it is protected by structures such as bark or bud scales, or is buried in duff or soil.

FIRE EFFECTS on plants

Fire effects on plants can vary significantly among fires and on different areas of the same fire.

Fire behavior, fire duration, the pattern of fuel consumption, and the amount of subsurface heating all influence injury and mortality of plants, and their subsequent recovery.

Postfire responses also depend upon the characteristics of the plant species on the site, their susceptibility to fire, and the means by which they recover after fire.

The **likelihood of plant tissue being killed by fire** depends upon the **amount of heat it receives**. The heat received by a plant is **determined by the temperature (T) reached and the duration of exposure**.

FIRE EFFECTS on plants

Some plant tissues, particularly growing points (meristems or buds) tend to be much more sensitive to heat when they are actively growing and their tissue moisture is high, than when their moisture content is low.

The concentration of other compounds that vary seasonally such as salts, sugars, and lignins may also be related to heat tolerance of plants.

Plant mortality depends on the amount of meristematic tissues killed. Susceptible tissue may not be exposed to heating by fire because it is protected by structures such as bark or bud scales, or is buried in duff or soil.

Plant mortality is often the result of injury to several different parts of the plant, such as **crown damage coupled with high cambial mortality.**

FIRE EFFECTS on plants

A woody plant's structure affects the probability that the aboveground portion will be killed by fire.

Important aerial crown characteristics include:

- branch density,
- ratio of live to dead crown material,
- location of the base of the crown with respect to surface fuels,
- total crown size.

Height enhances survival, as the aerial portions of small stature plants are almost always killed. Species of trees that self-prune their dead lower branches are less likely to have a fire carry into their crowns.

Small buds are more susceptible to lethal heating than large buds because of their small mass. The small diameter twigs and small buds of most shrub species make them fairly susceptible to fire.

For conifers, **long needles provide more initial protection to buds than short needles** that leave the bud directly exposed to heat from the fire. **Deciduous trees are much less susceptible during the dormant than growing season.**

FIRE EFFECTS on plants

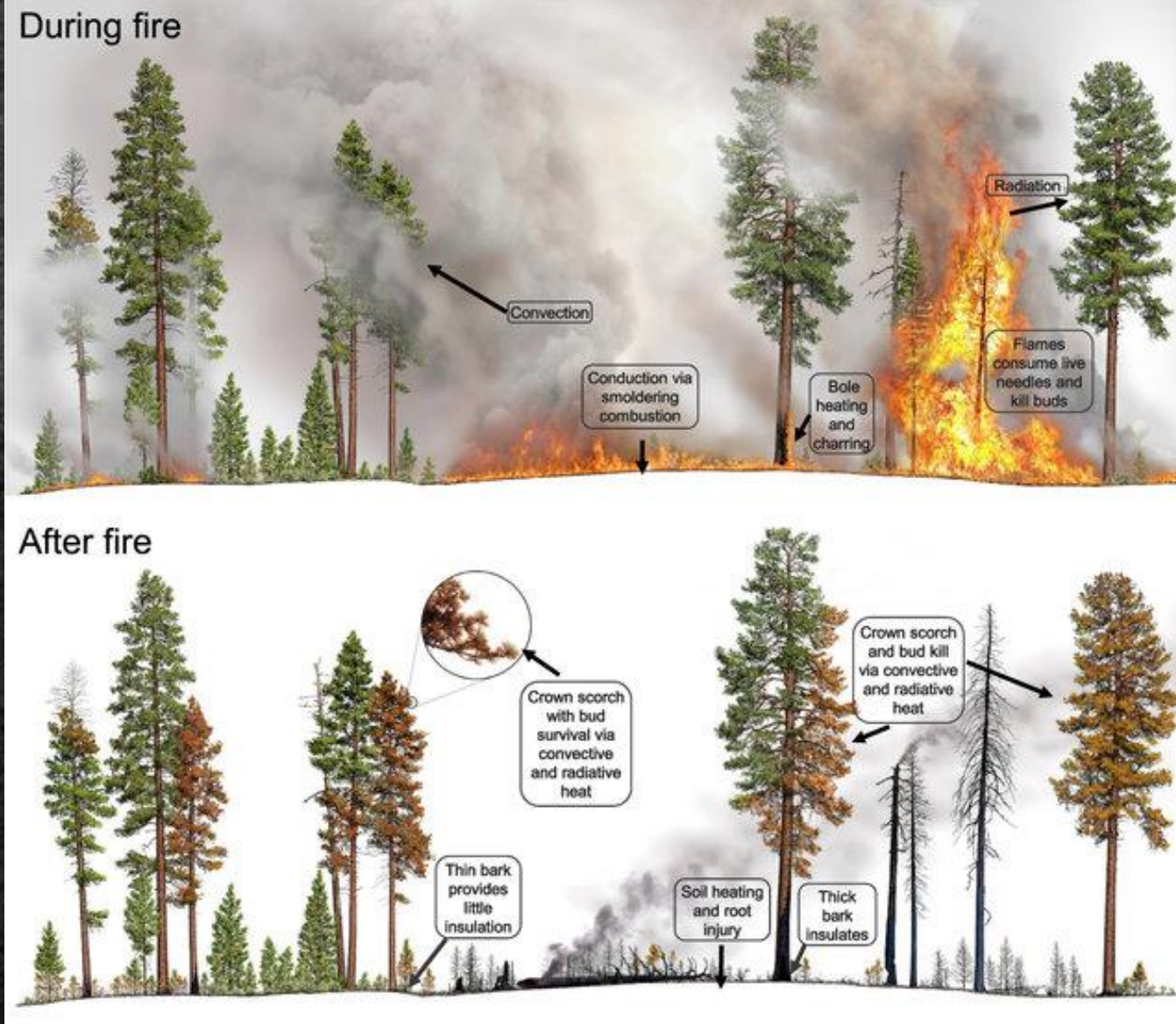
In fires where aerial crowns are not burned, **trees and shrubs can be killed by lethal heating of the cambial layer.**

Fire resistance of tree stems is most closely related to bark thickness, which varies with species, tree diameter and age, distance above the ground, site characteristics, and health and vigor of the tree.

Flame length, flaming residence time, and stem char height can be related to the amount of mortality of thin bark trees.

Cambium that grows beneath thick bark layers typically found on mature conifers is insulated from heat released by the flaming front. However, **the cambium can be killed by long-duration heating, such as from burnout of logs and smoldering combustion in deep litter and duff layers.**

FIRE EFFECTS on plants



Heat is transferred to living tissues of trees during fire (top panel), resulting in injuries to different parts of trees after fire (bottom panel). **Fire causes injuries to different parts of trees—buds, foliage, cambium in the stem, and roots**—through three different heat transfer processes.

Combustion directly consumes live foliage and buds, small live branches, and small trees and causes tissue death. Convection, the movement of hot air—and radiation, heat traveling as energy waves, causes tissue death when temperatures are $\geq 60^\circ\text{C}$ for 1 s.

Bole heating: Heat is conducted through the bark of trees, but because bark is a poor conductor it insulates the live cambium underneath from heat.

Thick bark insulates larger trees of some species, while thin bark provides little insulation on smaller trees and thin-barked species. Soil and root heating primarily occurs through conduction during smoldering combustion of duff and large logs.

FIRE EFFECTS on plants

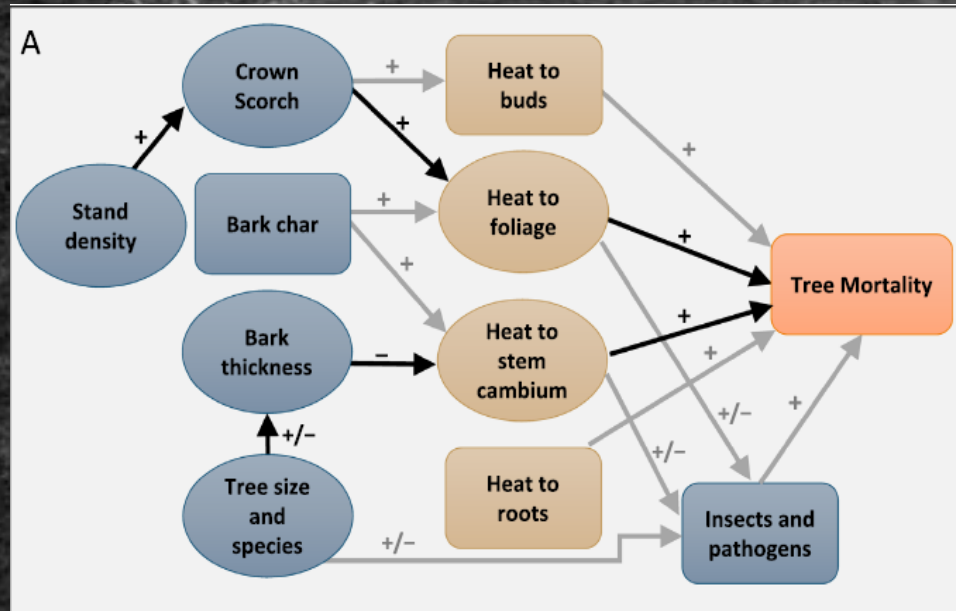
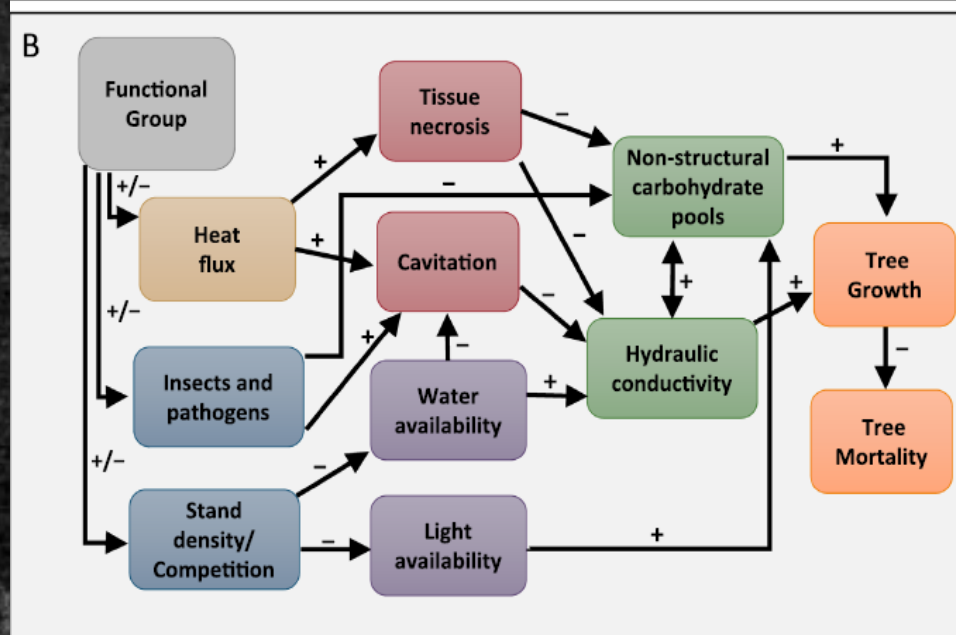


Figure 5. **(A) Conceptual model representing current modeling approaches to predicting post-fire tree mortality. Common inputs are shown in ovals. Additional, less-common inputs are shown in boxes.** Black arrows represent causal pathways that are usually included in the models, while gray arrows include pathways that are included less often. **No physiological processes are explicitly represented.**



(B) Conceptual model of the physiological processes that contribute to post-fire tree growth and mortality. Species are first grouped and assigned response functions to heat flux of living tissues (tan box), susceptibility to insects and disease, and competition tolerance (blue boxes). Short-term damage (red boxes) leads to long-term impairment of physiological status (green boxes), specifically the depletion of carbohydrate reserves or decreases in hydraulic function. Either of those factors, or their combined influence, can cause changes to tree growth and immediate or delayed tree mortality.

Exogenous factors (blue boxes) can directly cause damage, or alter the physical environment (purple boxes) leading to damage, impaired functioning, or improved functioning. Positive correlation (+), negative correlation (-), Species dependent (+/-).

FIRE EFFECTS on plants

In 2003, following a large wildfire in central Portugal, Catry et al. (2010) surveyed the area **during the first year and assessed the vegetative condition of 1040 burned trees from 11 different species**. Among those trees, **755 individuals were selected and monitored annually for 4 years**.

Almost all broadleaved trees survived, while most conifers died.

Despite of the low mortality observed in broadleaves, most were top-killed and regenerated only from basal resprouts, which implies a slow recovery process.

***Quercus suber* showed vigorous post-fire crown resprouting and was the most resilient species.**

Besides the differences between the two main functional groups (coniferous, broadleaved), **bole char height and crown volume scorched or consumed were important predictors of tree responses.**

The **main factor determining crown mortality on broadleaved species was bark thickness.**

FIRE EFFECTS on plants

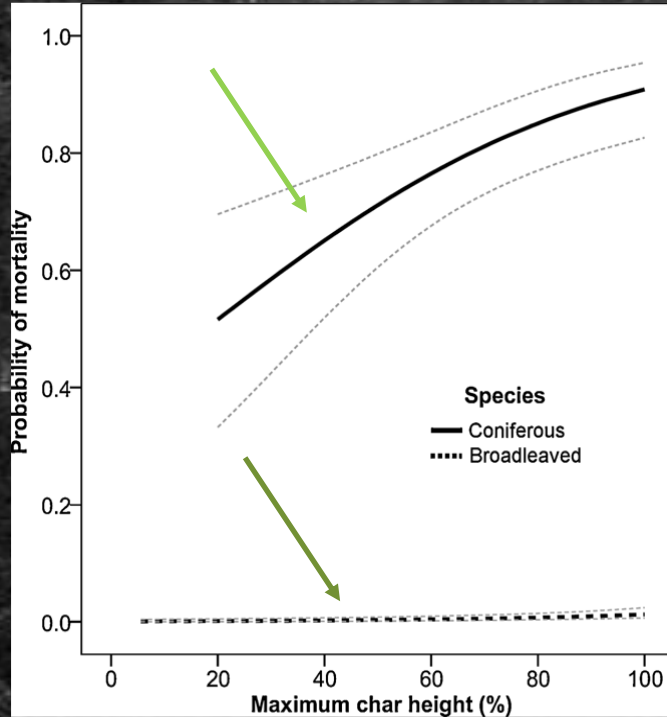


Fig. 2. **Probability of individual tree mortality** (with 95% confidence intervals) 1 year following wildfire **as a function of PCH (maximum bole char height expressed as percentage of tree height)** for the two species group considered (**coniferous vs. broadleaved trees**).

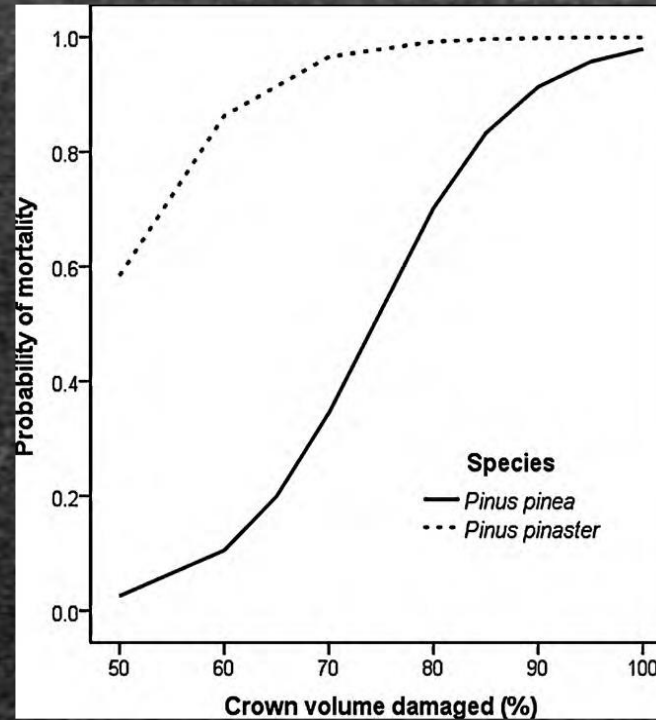


Fig. 3. **Probability of mortality for *P. pinea* and *P. pinaster*** 4 years following fire, **as a function of the percentage of crown volume damaged (TCD)**.

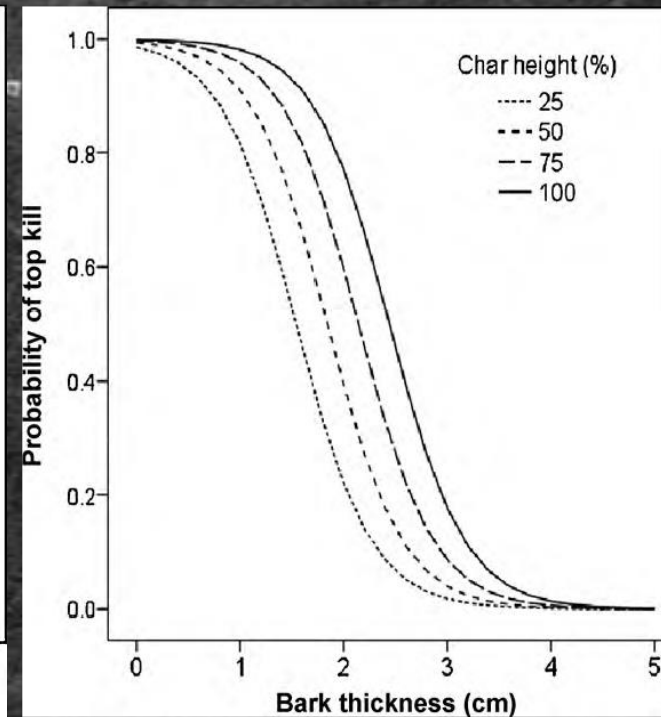


Fig. 4. **Probability of top-kill (crown mortality) following wildfire on broadleaved species, as a function of bark thickness (cm) and PCH maximum bole char height expressed as percentage of tree height**.

Empirical models based on fire effects data, usually not available.

Difficult to apply for management purposes.

FIRE EFFECTS on plants

Marques et al. (2011) developed **fire damage models for *Eucalyptus globulus* Labill stands in Portugal**. Models are **based on easily measurable forest characteristics**, so that forest managers may predict post-fire mortality based on forest structure.

A **three-step modelling strategy** based on logistic regression models to:

1. predict whether mortality occurs after a wildfire in a eucalypt stand.
2. estimate the degree of damage caused by wildfires in stands where mortality occurs.
3. distribute mortality among trees.

Data from **85 plots and 1648 trees** were used.

Relative damage increases with stand basal area.

Trees with large diameters, in **dominant positions** and located in **regular stands** are **less prone to die** when a wildfire occurs.

FIRE EFFECTS on plants

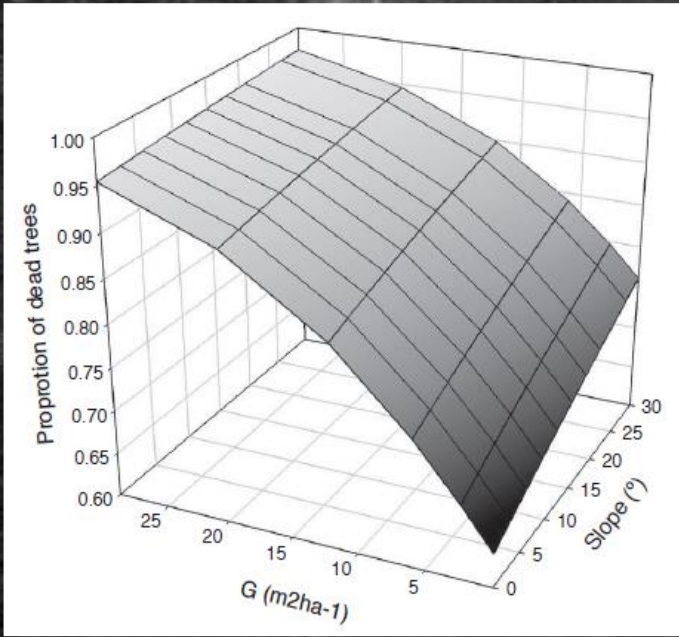


Fig. 2. **Effect of stand basal area (G) and slope on the degree of damage in the stand, i.e. the proportion of dead trees** (Eq. 6). The values were calculated with an altitude = 200 m and $S_d = 3.47$ cm.

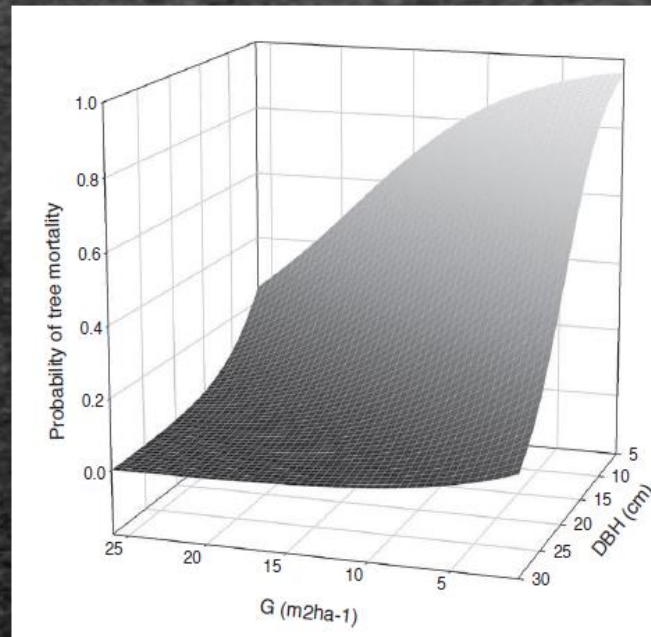


Fig. 4. **Effect of tree-diameter (dbh) and basal area (G) on the probability of tree mortality** using Eq. 7. The values were calculated using $Sh = 2$ which is the mean value in our dataset.

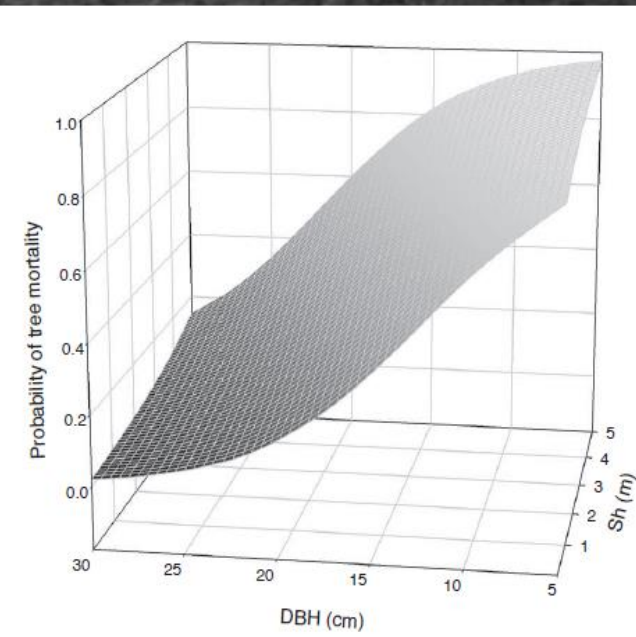


Fig. 5. **Effect of tree-diameter (DBH) and standard deviation of the tree height (Sh) on the probability of tree mortality** using Eq. 7. The values were calculated using $G = 6$ which is the mean value in our dataset.

Sh measures stand (ir)regularity.

Empirical models based on tree and stand level standard forest inventory data.

Easy to apply for management purposes.

FIRE EFFECTS on plants

Management actions contributing to **decrease fire intensity and severity** in *E. globulus* stands (e.g. reflected by a reduction of char height or char depth) will likely **increase the probability of tree and stem survival**.

Reducing vegetation cover, reducing the proportion of pine trees and increasing the proportion of other broadleaved trees may contribute to **decrease the probability of top-kill**.

Eucalypt resistance to fire was highly correlated with the thickness and the extent of protective bark tissue on the stem. These generally **increase with tree size**. If bark is sufficiently thick eucalyptus trees may even survive also crown fire.

Trees in stands with high tree height variability have higher probability of dying after fire because irregular structures may facilitate crown fires.

The combination of dbh and tree height variability indicates that **dominant trees** (e.g. trees with high diameters located in irregular stands) **have lower probability of dying**.

Catry, F. X., Moreira, F., Tujeira, R., & Silva, J. S. (2013). Post-fire survival and regeneration of *Eucalyptus globulus* in forest plantations in Portugal. *Forest Ecology and Management*, 310, 194-203.

Marques, S., Garcia-Gonzalo, J., Borges, J. G., Botequim, B., Oliveira, M. M., Tome, J., & Tomé, M. (2011). Developing post-fire *Eucalyptus globulus* stand damage and tree mortality models for enhanced forest planning in Portugal. *Silva Fennica*, 45(1): 69-83.

FIRE EFFECTS on plants

Mortality prediction models for mixed forests may be used for **selecting species for reforestation**, privileging those that are more fire-resistant. They may be used after the fire to **assess the probability of mortality or top-kill of individual trees** when developing **salvaging guidelines**, and to assess potential stand damages and economical losses when considering different management alternatives.

Mortality predictions can indicate **how exposed the burned areas will be to soil erosion**, or to **impacts on wildlife habitat**.

Ungulate herbivory may be an additional factor influencing the delayed mortality of top-killed broadleaves, because regular consumption of basal sprouts maintains low leaf area, hindering the adequate functioning of vital physiological processes.

For most forest species, the presence of either wild or domestic herbivores can have a negative impact on the post-fire recovery, even if not killing the trees, by damaging natural regrowth.